

CAPSTONE PROJECT REPORT



Air Quality Index Analysis Project

Using PostgreSQL, Python EDA, Power BI Dashboard, and PPT Reporting

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DATASET : AQI, Location, and Weather datasets for Indian cities

REPORT TYPE : End-to-end analytics and visualization project documentation

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I also extend my heartfelt thanks to **APPONIX Technologies, Bangalore**, for providing an excellent learning environment, quality training, and the necessary resources that enabled me to successfully undertake and complete this project. The practical exposure and industry-oriented approach offered by the organization greatly contributed to my understanding of real-world data analytics.

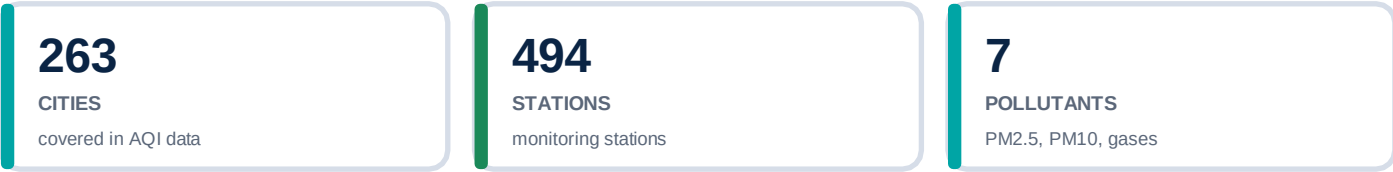
This project provided me with valuable hands-on experience in transforming raw environmental data into meaningful business insights through **database design using PostgreSQL, data analysis and exploratory data analysis (EDA) using Python, and interactive dashboard development using Power BI**. It significantly strengthened my understanding of the complete data analytics workflow from data collection and preprocessing to visualization and interpretation.

I would also like to acknowledge the importance of the **AQI, Location, and Weather** datasets used in this study. These datasets served as the foundation for analysing air pollution patterns across different states and cities while enabling a comprehensive examination of the relationship between air quality and key weather parameters such as **temperature, humidity, rainfall, wind speed and visibility**. Their availability made it possible to derive meaningful insights and practical recommendations for better environmental monitoring and decision-making.

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1. Introduction



Air Quality Index (AQI) analysis is an important method used to measure and monitor the quality of air in different regions. It helps us understand whether the air is clean, moderately polluted, or harmful to human health. Poor air quality can lead to serious health problems such as asthma, respiratory diseases, heart-related illnesses, and reduced overall well-being. It also affects daily activities, outdoor work, transportation, and the environment. Therefore, monitoring air quality is essential for protecting public health and supporting informed environmental decision-making.

The primary objective of this **AQI Analysis Project** is to analyse air pollution levels across different Indian states and cities and understand how various weather conditions influence air quality. The project combines air pollution data with weather information to identify patterns, trends, and relationships between environmental factors such as **temperature, humidity, rainfall, wind speed, and visibility** and different air pollutants. This analysis helps in gaining a better understanding of the factors that contribute to changes in air quality.

The project is based on three main datasets: **AQI Data, Location Data, and Weather Data**. These datasets are integrated to create a complete environmental analysis system. The AQI dataset contains pollution-related information for different cities, the Location dataset provides geographical details such as state and city names, and the Weather dataset contains weather parameters that can influence pollution levels. Together, these datasets provide a comprehensive view of environmental conditions across various locations.

A complete data analytics workflow has been followed throughout this project. The process begins with collecting and preparing the datasets, followed by data cleaning and preprocessing in Microsoft Excel to improve data quality. The cleaned datasets are then stored and managed in **PostgreSQL**, where database tables, relationships, and SQL queries are used to organize and retrieve the data efficiently. After database management, **Python** is used to perform **Exploratory Data Analysis (EDA)**, statistical analysis, and data visualization to discover meaningful patterns, identify trends, and generate useful insights. Finally, interactive dashboards are developed in **Power BI** to present the findings in a clear, visual, and easy-to-understand format, making the analysis more useful for decision-making.

This report presents every stage of the project in a structured and easy-to-understand manner. It explains the project objectives, datasets used, database design, SQL implementation, Python-based data analysis, visualizations, Power BI dashboards, key findings, recommendations, limitations, and future scope. The report demonstrates how raw environmental data can be transformed into meaningful insights that support better environmental monitoring, pollution management, and informed policy decisions.

2. Problem Statement and Objectives

Air pollution has become one of the most significant environmental and public health challenges in recent years. Every day, large volumes of air quality and weather data are collected from different monitoring stations across various cities and states. However, this data is usually available in its raw form, making it difficult to identify important trends, compare pollution levels, or understand the factors affecting air quality. Raw data alone cannot provide meaningful answers unless it is properly organized, cleaned, analysed, and presented in a clear and understandable manner.

The main problem addressed in this project is to transform raw **Air Quality Index (AQI)**, **Location**, and **Weather** datasets into meaningful and actionable insights. The project aims to analyse pollution levels across different regions of India and study how various environmental factors influence air quality. By integrating pollution data with weather information, the analysis helps identify patterns, relationships, and trends that are not immediately visible from the raw datasets.

Another important challenge is to manage and analyse the data efficiently using modern data analytics tools. This project demonstrates how environmental data can be stored in a structured database, analysed using SQL and Python, and finally presented through interactive Power BI dashboards. The end result is a user-friendly analytical solution that allows stakeholders to better understand pollution conditions and make informed environmental decisions.

Project Objectives

The primary objectives of this AQI Analysis Project are:

- To calculate and compare the **average pollutant values** across different states and cities to identify areas with higher pollution levels.
- To identify the **pollutants with the highest average and maximum concentration**, helping determine which pollutants contribute most significantly to poor air quality.
- To analyse the relationship between **weather parameters** such as **temperature, humidity, rainfall, wind speed, visibility, and UV index** and air pollution levels to understand their impact on AQI.
- To clean, organize, and store the datasets efficiently in **PostgreSQL** by creating well-structured database tables and performing SQL-based data analysis.
- To perform **Exploratory Data Analysis (EDA)** using **Python**, including statistical analysis and visualizations that reveal hidden trends, distributions, correlations, and patterns within the data.
- To design interactive and informative **Power BI dashboards** that present key performance indicators (KPIs), charts, filters, and visual insights in an easy-to-understand format.
- To summarize the complete project through a professional **project report and presentation (PPT)** that clearly communicates the methodology, analysis, findings, recommendations, limitations, and overall outcomes of the study.

Overall, the objective of this project is to demonstrate how raw environmental data can be transformed into meaningful information through the complete data analytics process, enabling better monitoring of air quality, supporting data-driven decision-making, and contributing to improved environmental awareness and pollution management.

3. Understanding AQI

The Air Quality Index (AQI) is a standardized measurement used to indicate how clean or polluted the air is at a particular location. It converts complex air pollution measurements into a simple numerical scale that is easy for the public to understand. AQI helps people determine whether the air is safe for outdoor activities and provides an indication of the possible health risks associated with different pollution levels.

A lower AQI value represents cleaner air with minimal health concerns, while a higher AQI value indicates increasing levels of air pollution and greater health risks. As pollution levels rise, the likelihood of respiratory problems, heart-related illnesses, eye irritation, and other health issues also increases, particularly for children, older adults, and individuals with existing medical conditions. Government agencies and environmental organizations use AQI to monitor air quality, issue public health advisories, and support pollution control measures.

In this project, the primary measurement used for analysis is the Average Pollutant Value, which represents the average concentration of different pollutants recorded at various monitoring locations. In addition to the average values, the dataset also contains Minimum Pollutant Values and Maximum Pollutant Values. These measurements help compare pollution levels across different cities, states, monitoring stations, and pollutant types while identifying areas with consistently high or low pollution levels.

The AQI dataset also includes information about different pollutant categories, allowing the analysis to identify which pollutants contribute most significantly to poor air quality. By combining AQI data with weather information such as temperature, humidity, rainfall, wind speed, visibility, and UV index, the project provides a more comprehensive understanding of the environmental conditions that influence pollution levels.

Understanding these AQI categories is essential because they help governments, environmental agencies, healthcare professionals, and the public assess pollution levels quickly and take appropriate preventive measures. In this project, these AQI classifications are used to interpret pollution patterns, compare air quality across different locations, and generate meaningful insights through statistical analysis and interactive dashboards.

The following table explains the commonly used AQI categories and their general meaning:

AQI Range	Category	Simple Meaning	Possible Impact
0 - 50	Good	Air is clean	Little or no health risk
51 - 100	Satisfactory / Moderate	Acceptable air quality	Sensitive people may feel mild effects
101 - 200	Moderate / Poor	Pollution is noticeable	People with asthma or breathing problems should limit prolonged outdoor activities.
201 - 300	Poor / Very Poor	Air is unhealthy	Increased risk of respiratory and cardiovascular problems; outdoor exposure should be reduced.
Above 300	Severe	Very polluted air	Serious health risks for the general population; health warnings and strong pollution control measures are required.

4. Dataset Overview

3,288

AQI ROWS

pollutant records

267

LOCATION ROWS

city master records

263

WEATHER ROWS

weather records

The success of any data analytics project depends on the quality and organization of the data being analysed. This AQI Analysis Project is built using a well-structured collection of environmental datasets that provide information about air pollution, geographical locations, and weather conditions across different cities and states in India. Together, these datasets enable a comprehensive analysis of air quality and its relationship with various weather parameters.

The complete project package includes several files used throughout different stages of the analytics workflow. These include cleaned CSV datasets, a combined Microsoft Excel workbook, PostgreSQL SQL scripts for database creation and querying, a Python Jupyter Notebook for Exploratory Data Analysis (EDA), Power BI dashboard files for interactive visualization, and a PowerPoint presentation summarizing the project's objectives, methodology, analysis, and findings.

The project primarily uses three core datasets: Location, AQI, and Weather. These datasets are interconnected to provide a complete environmental analysis system. The Location dataset acts as the master reference table by storing geographical information such as states and cities. The AQI dataset contains pollution-related measurements collected from monitoring stations, while the Weather dataset provides meteorological information that helps analyse how weather conditions influence air pollution levels.

It is important to note that the row counts shown in the project presentation include the header row. However, after importing the datasets into Python using the Pandas library, the actual number of data records available for analysis is 3,288 rows in the AQI dataset, 267 rows in the Location dataset, and 263 rows in the Weather dataset.

The following table summarizes the datasets used in this project:

Dataset	Rows Loaded in Python	Columns	Purpose
AQI DATASET.csv	3,288	13	Main pollution table with pollutant values by station, city, and state
LOCATION DATASET.csv	267	3	Master table for state and city information
WEATHER DATASET.csv	263	18	Weather table with temperature, humidity, rainfall, wind, visibility, and UV index
AQI ANALYSIS DATASET.xlsx	3 sheets	Location, AQI, Weather	Combined Excel workbook used for project organization

5. Location Dataset

30

STATES

in location master

263

CITIES

unique city names

0

DUPLICATES

duplicate rows found

The Location Dataset serves as the master reference table for the entire AQI Analysis Project. It contains the geographical information required to identify the locations where air quality and weather observations have been recorded. Instead of repeatedly storing the same state and city names in multiple datasets, this table stores each location only once, making the database more organized, efficient, and easier to maintain.

The primary purpose of the Location dataset is to establish a common reference that links the different datasets used in the project. Each location is assigned a unique Location_ID, which acts as the Primary Key (PK) of the table. This unique identifier is then used as a Foreign Key (FK) in both the AQI and Weather tables, creating relationships between the datasets and ensuring data consistency.

Using a separate master table for location information follows the principles of database normalization. This approach minimizes data duplication, reduces storage requirements, prevents inconsistencies during data updates, and improves overall database performance. It also simplifies SQL queries because all location-related information can be retrieved through table relationships instead of storing duplicate city and state names in multiple tables.

The Location dataset plays a crucial role throughout the project. It enables pollution and weather records to be grouped, filtered, and compared across different states and cities, making regional analysis more accurate and efficient. It also supports the creation of relationships in PostgreSQL and Power BI, allowing interactive dashboards and analytical reports to display location-based insights effectively.

The table contains the following columns:

Column	Meaning
Location_ID	A unique numerical identifier assigned to each location. It serves as the Primary Key of the table and is used to establish relationships with the AQI and Weather datasets.
state	The name of the Indian state or union territory where the monitoring location is situated. This column is used for state-wise pollution analysis and comparisons.
city	The name of the city corresponding to the monitoring location. This column enables city-level analysis of air quality and weather conditions.

Overall, the Location dataset acts as the backbone of the database structure by providing a centralized and standardized source of geographical information. It ensures that pollution and weather data remain properly connected, making the database more scalable, easier to manage, and suitable for performing accurate environmental analysis across different regions of India.

6. AQI Dataset

The **AQI Dataset** is the primary dataset used in this project and forms the foundation of the entire air quality analysis. It contains detailed information about air pollutant measurements collected from various monitoring stations located across different cities and states in India. Each record in the dataset represents the concentration of a specific pollutant measured at a particular monitoring station, making it possible to analyse pollution levels at both local and regional scales.

The dataset includes important pollution measurements such as the **minimum, average, and maximum pollutant values**, which help evaluate variations in air quality over different locations. These measurements are used throughout the project to calculate summary statistics, identify highly polluted areas, compare pollution levels between cities and states, and generate visualizations for better interpretation of the data.

A total of **seven major air pollutants** are included in the dataset: **PM2.5, PM10, NO₂, SO₂, CO, Ozone (O₃), and NH₃**. These pollutants are routinely monitored because they have a significant impact on environmental quality and human health. Fine particulate matter such as **PM2.5** and **PM10** can enter the respiratory system and cause breathing problems, while gases such as **NO₂, SO₂, CO, Ozone**, and **NH₃** can contribute to respiratory illnesses, reduced visibility, environmental damage, and other health-related concerns. Monitoring these pollutants helps identify pollution sources and supports effective environmental management.

The AQI dataset is used extensively throughout the project for SQL-based analysis, Python Exploratory Data Analysis (EDA), statistical calculations, and Power BI dashboard creation. It enables the identification of pollution trends, comparison of pollutant concentrations across different regions, and analysis of pollutant distributions by city, state, monitoring station, and pollutant type.

The overall summary of the AQI dataset is shown below:

Metric	Value
Rows	3,288
Columns	13
States	29
Cities	263
Monitoring Stations	494
Average pollutant value	30.18
Maximum pollutant max value	500
Maximum pollutant avg value	312

7. Weather Dataset

The **Weather Dataset** provides the environmental and meteorological information used in this project to study how different weather conditions influence air quality. While the AQI dataset measures pollution levels, the Weather dataset helps explain the environmental factors that may contribute to changes in those pollution levels. By combining both datasets, the project performs a more comprehensive analysis of the relationship between weather and air pollution.

Weather conditions play an important role in determining how pollutants behave in the atmosphere. For example, **strong winds** can disperse pollutants and improve air quality by carrying them away from an area, whereas **low wind speeds** may allow pollutants to accumulate near the ground. **Rainfall** helps remove dust and harmful particles from the air through a natural cleaning process known as *wet deposition*. **Humidity** can influence the formation and suspension of fine particulate matter, while **temperature** affects chemical reactions in the atmosphere that contribute to the formation of secondary pollutants such as ozone. Similarly, **visibility** often decreases when pollution levels are high because airborne particles scatter and absorb light.

The Weather dataset contains several important environmental measurements, including **average temperature, humidity, rainfall, wind speed, atmospheric pressure, visibility, cloud cover, and UV index**. These variables are analysed together with AQI data to identify possible relationships between weather conditions and pollution levels across different cities and states.

To integrate weather information with air quality measurements, the Weather dataset is linked to the **AQI** and **Location** datasets primarily through **Location_ID**, along with corresponding **city** and **state** information. This relationship enables location-wise analysis and helps compare pollution levels under different weather conditions.

The summary statistics for the major weather parameters used in this project are presented below:

Weather Metric	Average	Minimum	Maximum
Temperature avg in C	32.35	7.70	39.60
Humidity Pct	38.67	15.40	94.80
Rainfall in mm	1.16	0.00	15.90
Wind speed in km/h	20.95	8.50	34.80
Visibility in km	10.96	3.00	19.70
UV index	8.51	3.80	11.60

8. Database Design and Relationship

A well-designed database is essential for storing, organizing, and retrieving data efficiently. In this project, a **relational database model** has been used because it provides a structured way to manage large volumes of environmental data while maintaining consistency and reducing unnecessary duplication. The database is designed using three interconnected tables: **Location_Master**, **AQI_Data**, and **Weather_Data**. Each table has a specific purpose, and together they form the foundation of the project's data management system.

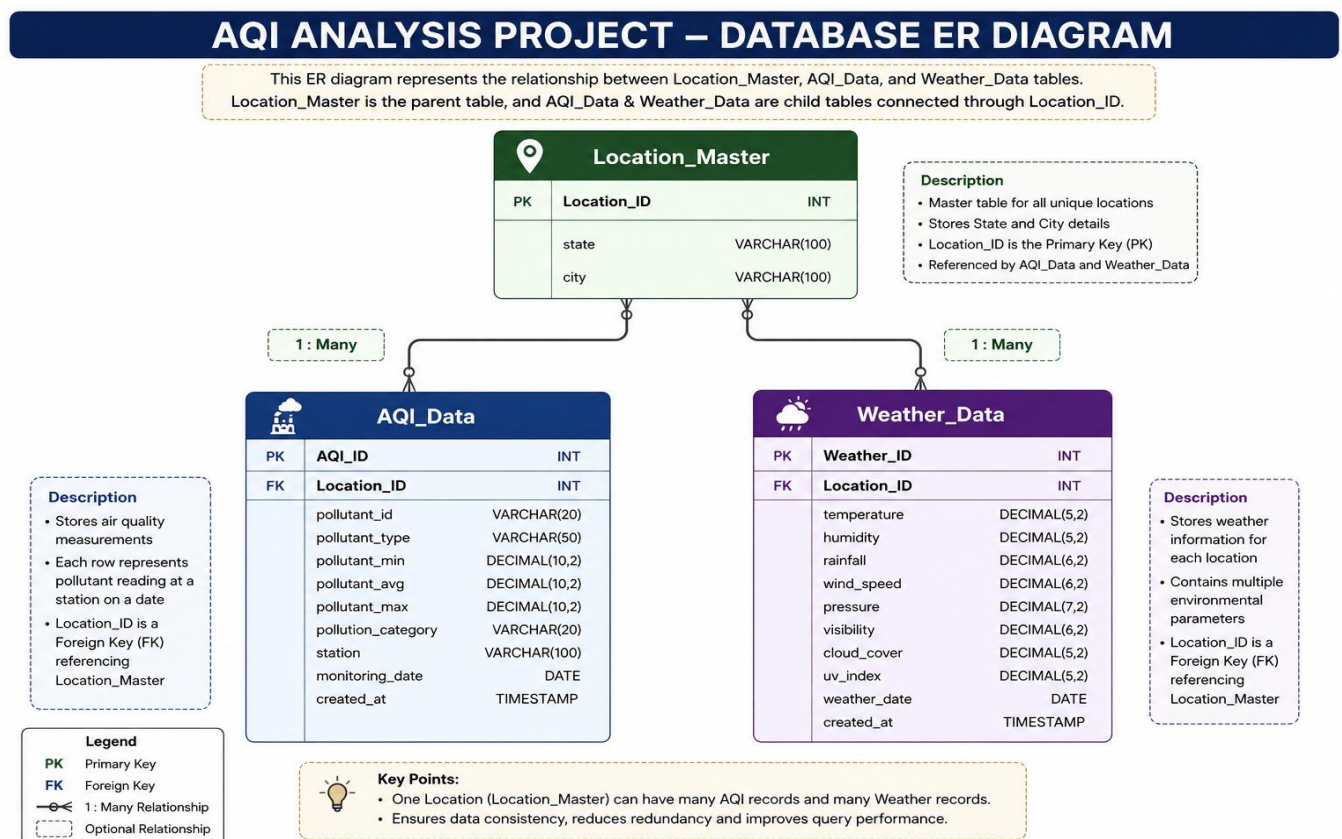
The **Location_Master** table acts as the **parent (master) table** because it stores the unique geographical information for each location, including the state and city names. Every location is assigned a unique **Location_ID**, which serves as the **Primary Key (PK)** of the table. This unique identifier ensures that each location is represented only once in the database.

The **AQI_Data** and **Weather_Data** tables are the **child tables**. Both tables contain a **Location_ID** column, which acts as a **Foreign Key (FK)** and links each record back to the corresponding location stored in the **Location_Master** table. This relationship allows pollution and weather data to be associated with the correct city and state without repeatedly storing the same geographical information in multiple tables.

This database structure follows the principles of **database normalization**, where common information is stored only once and shared across related tables. Such a design minimizes data redundancy, improves storage efficiency, maintains data integrity, and makes future updates much easier. For example, if the name of a city or state needs to be corrected, the change only needs to be made once in the **Location_Master** table, and the updated information automatically applies to all related AQI and Weather records through the database relationship.

The relational model also simplifies SQL operations such as **JOINS**, allowing information from multiple tables to be combined for analysis. This makes it possible to retrieve pollution measurements together with weather conditions and geographical details in a single query. The same relationships are later used in **Power BI** to create interactive dashboards and perform location-based filtering and analysis.

The figure below illustrates the database relationship used in this project.



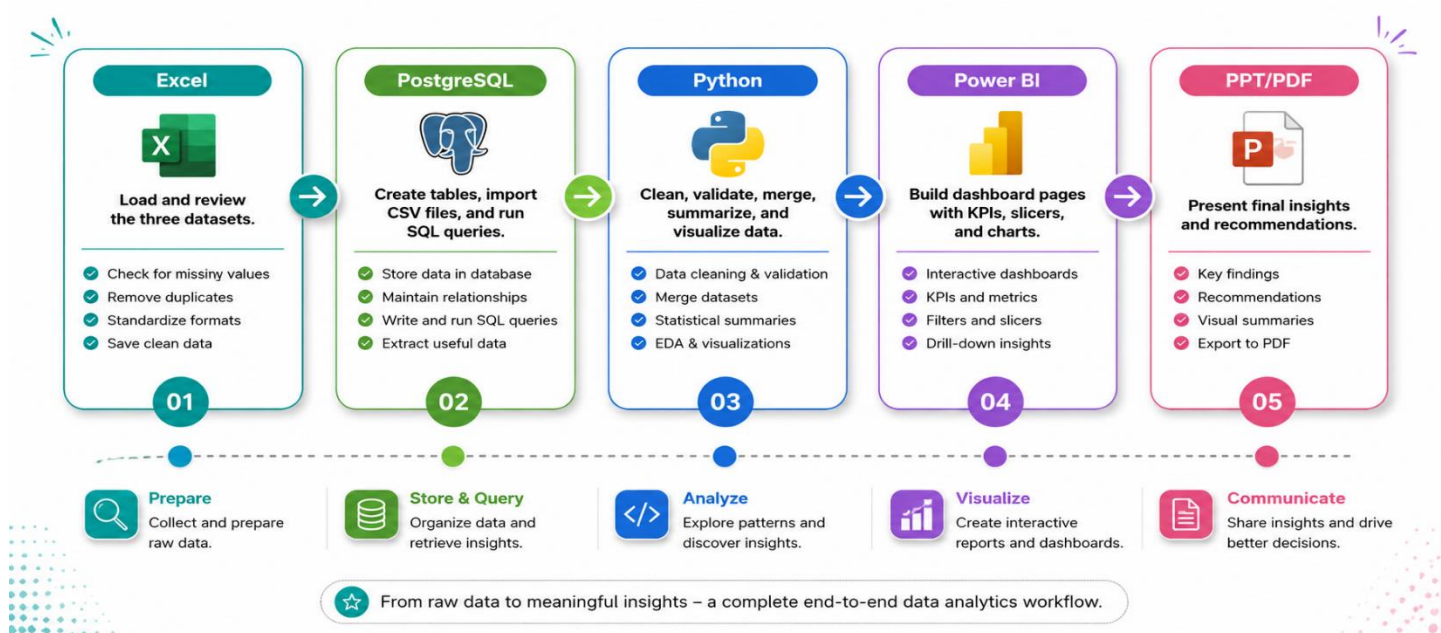
9. Project Workflow and Tools

The Air Quality Index (AQI) Analysis project follows a complete **end-to-end data analytics workflow**, where each tool is used for a specific purpose. The process starts with collecting and checking the data, then storing it in a database, analysing it, creating visual dashboards, and finally preparing a professional report. Using different tools at different stages makes the analysis more organized, accurate, and easy to understand.

- **Microsoft Excel** is used for the initial data collection, checking, and cleaning process. Missing values, duplicate records, and formatting issues are identified and corrected before moving the data to the database.
- **PostgreSQL** is used to store the cleaned datasets in relational tables. SQL queries help retrieve, filter, join, and analyse the data efficiently while maintaining relationships between the tables.
- **Python** is used to perform **Exploratory Data Analysis (EDA)**. Different statistical summaries, graphs, and visualizations are created to understand pollution patterns, weather conditions, and relationships between different variables.
- **Power BI** is used to build interactive dashboards. It converts the analysed data into charts, KPIs, maps, and filters that make it easy for users to explore the insights and compare pollution levels across cities, states, and pollutants.
- **Microsoft PowerPoint** is used to present the complete project in a simple and professional way. It explains the objectives, methodology, dashboards, key findings, recommendations, and conclusions clearly for the audience.

Key Benefits of the Workflow

- The workflow is **beginner-friendly** because every step follows a logical sequence, making the project easy to understand and execute.
- Each tool performs a different task, so the work becomes more organized and efficient instead of doing everything in a single application.
- The same data is checked and validated at multiple stages, which helps reduce errors and improves the accuracy of the final results.
- The dashboards provide clear visual insights that help users quickly identify pollution trends, weather patterns, and environmental conditions.
- The final **PDF project report** combines all the project details, analysis, dashboards, and recommendations into a single document, making it easy to review, present, and share with others.



10. Data Cleaning and Preprocessing

Data cleaning and preprocessing is one of the most important steps in any data analytics project because accurate data leads to reliable results. Before starting the analysis, all three datasets were carefully checked and prepared to ensure that the information was complete, consistent, and ready for further processing. This helped reduce errors and improved the overall quality of the analysis.

- The datasets were first examined in **Microsoft Excel** to identify formatting issues, incorrect values, and inconsistencies. Basic cleaning was performed to ensure that the data was properly organized.
- The cleaned CSV files were then validated using **Python (Pandas)**. The datasets were checked for **missing values (NULL values)**, **duplicate records**, incorrect **data types**, and inconsistent data entries. After validation, it was confirmed that **no missing values and no duplicate rows** were present in any of the three datasets.
- The data types of all columns were verified to ensure that dates, numerical values, and text fields were stored in the correct format. This prevents calculation errors and ensures that charts and statistical analysis produce accurate results.
- A **Location Master** table was created to provide a unique **Location_ID** for each city. This key was used to establish relationships between the AQI and Weather datasets, making the database more organized and reducing repeated information.
- The AQI and Weather datasets both contain observations for the date **31-05-2026**. Since all records belong to the same date, this project performs a **cross-sectional analysis**, which compares air quality and weather conditions across different cities and monitoring stations on that particular day, rather than analyzing changes over a long period of time.
- After completing all the cleaning and validation steps, the datasets were considered accurate, consistent, and ready for **database management in PostgreSQL**, **Exploratory Data Analysis (EDA) in Python**, and **interactive dashboard creation in Power BI**.

Outcome of Data Cleaning

- ✓ No missing (NULL) values were found in any dataset.
- ✓ No duplicate rows were detected after validation.
- ✓ Data types were verified and corrected where necessary.
- ✓ Relationships between tables were created using **Location_ID**.
- ✓ The datasets were successfully prepared for analysis and visualization, ensuring accurate and trustworthy insights.

Check	AQI Dataset	Location Dataset	Weather Dataset
Missing values	0	0	0
Duplicate rows	0	0	0
Main ID	AQI_ID	Location_ID	Weather_ID
Relationship key	Location_ID	Location_ID	Location_ID
Date coverage	31-05-2026	Not date based	31-05-2026

11. PostgreSQL Implementation

After cleaning the datasets, they were imported into **PostgreSQL** for structured storage and efficient data management. PostgreSQL serves as the central database of the project, where the CSV files are converted into relational tables. Using a database instead of Excel makes the data easier to organize, retrieve, update, and analyse, especially when working with multiple datasets.

The database consists of **one master table (Location_Master)** and **two transaction/detail tables (AQI_Data and Weather_Data)**. The **Location_Master** table stores the unique details of each city and state, while the **AQI_Data** and **Weather_Data** tables store pollution and weather information. Both tables contain a **Location_ID** as a **Foreign Key**, which is linked to the **Primary Key** in the **Location_Master** table. This relationship helps avoid duplicate location information and maintains data consistency across the database.

SQL scripts were used to create the tables, define **Primary Keys** and **Foreign Keys**, and import the cleaned CSV files into PostgreSQL. Once the data was loaded, several SQL queries were executed to verify that the records had been imported correctly and that the relationships between the tables were working as expected.

The relational database structure allows data from different tables to be combined using **SQL JOIN operations**. This makes it possible to analyse air quality and weather information together, providing more meaningful insights than analysing each dataset separately.

Benefits of Using PostgreSQL

- The cleaned CSV files are stored in a structured and organized relational database.
- Primary Key and Foreign Key relationships maintain data accuracy and prevent duplicate location records.
- SQL queries can quickly retrieve, filter, sort, and summarize large amounts of data.
- Multiple tables can be connected using **JOIN** operations to perform combined analysis.
- The database serves as a reliable source for Python EDA and Power BI dashboards.

Table 1: location_master



```
CREATE TABLE location_master (  
  location_id VARCHAR(10) PRIMARY KEY,  
  state VARCHAR(100),  
  city VARCHAR(100)  
);
```

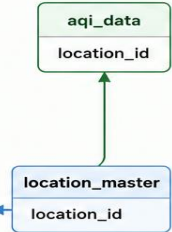
Column Name	Data Type	Constraint
location_id	VARCHAR(10)	PRIMARY KEY
state	VARCHAR(100)	
city	VARCHAR(100)	

Table 2: aqi_data



```
CREATE TABLE aqi_data (  
  aqi_id VARCHAR(10) PRIMARY KEY,  
  location_id VARCHAR(10),  
  country VARCHAR(50),  
  date DATE,  
  state VARCHAR(100),  
  city VARCHAR(100),  
  station TEXT,  
  pollutant_id VARCHAR(50),  
  pollutant_min INT,  
  pollutant_max INT,  
  pollutant_avg INT,  
  
  FOREIGN KEY (location_id) REFERENCES location_master(location_id)  
);
```

Column Name	Data Type	Constraint
aqi_id	VARCHAR(10)	PRIMARY KEY
location_id	VARCHAR(10)	FOREIGN KEY
country	VARCHAR(50)	
date	DATE	
state	VARCHAR(100)	
city	VARCHAR(100)	
station	TEXT	
pollutant_id	VARCHAR(50)	
pollutant_min	INT	
pollutant_max	INT	
pollutant_avg	INT	



12. SQL Queries and Analytical Use

After the data was stored in PostgreSQL, SQL queries were used to validate the datasets and answer important project questions. SQL provides a fast and efficient way to retrieve, filter, group, and summarize data directly from the database. These queries also helped verify that the imported data was correct before moving to further analysis in Python and visualization in Power BI.

Simple SQL statements were written to perform different types of analysis on the AQI and Weather datasets. The results obtained from these queries served as the foundation for the Exploratory Data Analysis (EDA) and dashboard creation stages. By checking the data in SQL first, the chances of errors in later stages were greatly reduced.

Some of the key SQL queries performed in this project include:

- Counting the total number of rows in each table to verify successful data import.
- Displaying the list of distinct states and cities available in the datasets.
- Calculating the **average pollutant value** for each city to identify areas with higher pollution levels.
- Counting the number of records for each **pollutant type** such as PM2.5, PM10, NO₂, CO, and Ozone.
- Filtering records where the **average pollutant value is greater than 100** to identify cities with poor air quality.
- Sorting and grouping data to compare pollution levels across different locations.
- Joining AQI and Weather tables using **Location_ID** to analyse pollution and weather information together.

Importance of SQL in the Project

- SQL allows data to be retrieved and summarized quickly without processing the entire dataset.
- It helps validate the accuracy of imported data before performing advanced analysis.
- Grouping, filtering, and aggregation functions make it easy to identify important trends and patterns.
- SQL queries provide clean datasets that can be directly used in Python and Power BI.

QUERY 1

```
SELECT city, AVG(pollutant_avg) AS avg_aqi
FROM aqi_data
GROUP BY city
ORDER BY avg_aqi DESC;
```

QUERY 2

```
SELECT pollutant_id, COUNT(*) AS pollutant_count FROM aqi_data
GROUP BY pollutant_id
ORDER BY pollutant_count DESC;
```

QUERY 3

```
SELECT city, pollutant_id, pollutant_avg
FROM aqi_data
WHERE pollutant_avg > 100
ORDER BY pollutant_avg DESC;
```

13. Python EDA Notebook Structure

The **Jupyter Notebook** is the main analysis component of this project. It is used to perform **Exploratory Data Analysis (EDA)** on the AQI, Weather, and Location datasets. The notebook follows a step-by-step approach, beginning with importing the required Python libraries, loading the datasets, checking the data quality, performing statistical analysis, and finally creating visualizations to discover meaningful patterns and insights.

The notebook is designed in a simple and organized manner so that each stage of the analysis can be easily understood. Every code section is accompanied by explanations and visual outputs, making the workflow suitable for beginners as well as project presentations.

The notebook uses four important Python libraries:

- **Pandas** is used to load the CSV files, inspect the data, clean missing or duplicate values, merge datasets, group records, calculate summary statistics, and prepare the data for analysis.
- **NumPy** is used to perform numerical and mathematical operations efficiently. It supports calculations such as averages, minimum and maximum values, and other statistical computations.
- **Matplotlib** is used to create basic data visualizations including bar charts, histograms, pie charts, line charts, and scatter plots. These charts help present the data in an easy-to-understand visual format.
- **Seaborn** is built on top of Matplotlib and is used to create more advanced statistical visualizations such as box plots, violin plots, heatmaps, and enhanced scatter plots. These charts make it easier to identify trends, distributions, correlations, and outliers.

The notebook contains a variety of charts to analyse different aspects of the data, including:

- **Bar Charts** to compare average pollution levels across cities, states, and pollutant types.
- **Histograms** to understand the distribution of temperature, humidity, rainfall, and pollutant values.
- **Pie Charts** to show the proportion of different pollution categories or pollutant types.
- **Scatter Plots** to examine relationships between pollution levels and weather variables such as temperature, humidity, and rainfall.
- **Box Plots** to identify the spread of pollutant values and detect outliers.
- **Violin Plots** to visualize the distribution of weather variables across different pollutant categories.
- **Line Charts** to compare average pollutant values for different pollutant types in a clear and simple way.

These visualizations help answer important questions such as:

- Which cities and states have the highest pollution levels?
- Which pollutants are most common?
- How are weather conditions related to air pollution?
- How are pollutant values distributed across different locations?
- Are there any unusual or extreme pollution readings?

The complete notebook contains **130 cells**, including **46 code cells** and the remaining markdown cells for explanations, observations, and insights. This combination of code, charts, and written explanations makes the notebook easy to follow, well-documented, and suitable for both learning and professional project presentations.

14. EDA Data Quality Checks

Before creating any charts or performing statistical analysis, it is important to verify that the datasets are accurate and complete. Data quality checks help identify problems such as missing values, duplicate records, incorrect data types, and inconsistent data formats. If these issues are not detected early, they can lead to incorrect calculations, misleading visualizations, and unreliable conclusions.

In this project, all three datasets—**AQI**, **Location**, and **Weather**—were carefully examined using Python and the **Pandas** library before starting the Exploratory Data Analysis (EDA). Several validation steps were performed to ensure that the data was clean, consistent, and ready for analysis.

The following checks were carried out:

- The **shape of each dataset** was verified to confirm the total number of rows and columns.
- The **column names** were inspected to ensure that all required fields were present and correctly labeled.
- The **data types** of every column were checked so that numbers, dates, and text values were stored in the correct format.
- The datasets were scanned for **missing (NULL) values** to make sure no important information was absent.
- A **duplicate row check** was performed to confirm that no records were repeated, preventing the same data from being counted multiple times.
- The datasets were also reviewed for overall consistency before merging and visualization.

After completing these validation steps, it was confirmed that the datasets were clean and suitable for further analysis. **No missing values** and **no duplicate rows** were found in any of the three core CSV files. This ensured that the statistical calculations, charts, and dashboard insights generated later in the project were based on accurate and reliable data.

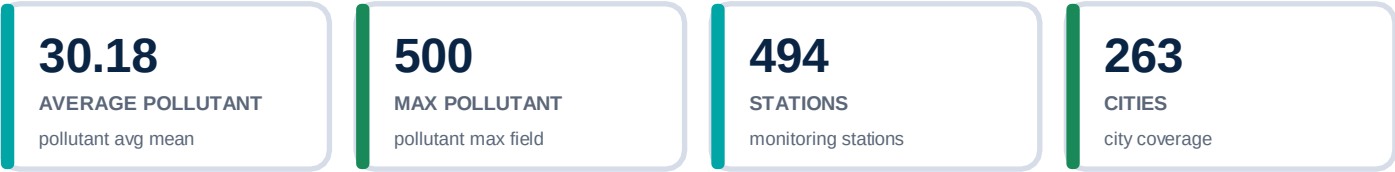
Summary of Data Quality Checks

Dataset	Rows	Columns	Missing Cells	Duplicate Rows
AQI	3,288	13	0	0
Location	267	3	0	0
Weather	263	18	0	0

Key Observations

- ✓ All three datasets were successfully loaded into Python.
- ✓ The dataset structure and column names were verified.
- ✓ All data types were checked and found to be appropriate for analysis.
- ✓ No missing (NULL) values were present.
- ✓ No duplicate rows were detected.
- ✓ The datasets were clean, consistent, and ready for Exploratory Data Analysis (EDA), SQL querying, and Power BI dashboard development.

15. KPI Baseline from the Data



Before creating the Power BI dashboards, several **Key Performance Indicators (KPIs)** were calculated to provide a quick summary of the overall air quality and weather conditions in the dataset. These KPIs act as baseline values, helping users understand the general environmental conditions at a glance without exploring the detailed data.

The **average pollutant value** across all AQI records is **30.18**, which represents the overall pollution level observed in the dataset. This KPI provides a benchmark for comparing pollution levels across different cities, states, and pollutant types. The dashboard also displays the **maximum pollutant value**, which is **500**. This value is taken from the **pollutant_max** column and represents the highest pollution reading recorded in the dataset. Such extreme values indicate locations where air quality reached hazardous levels and require immediate attention.

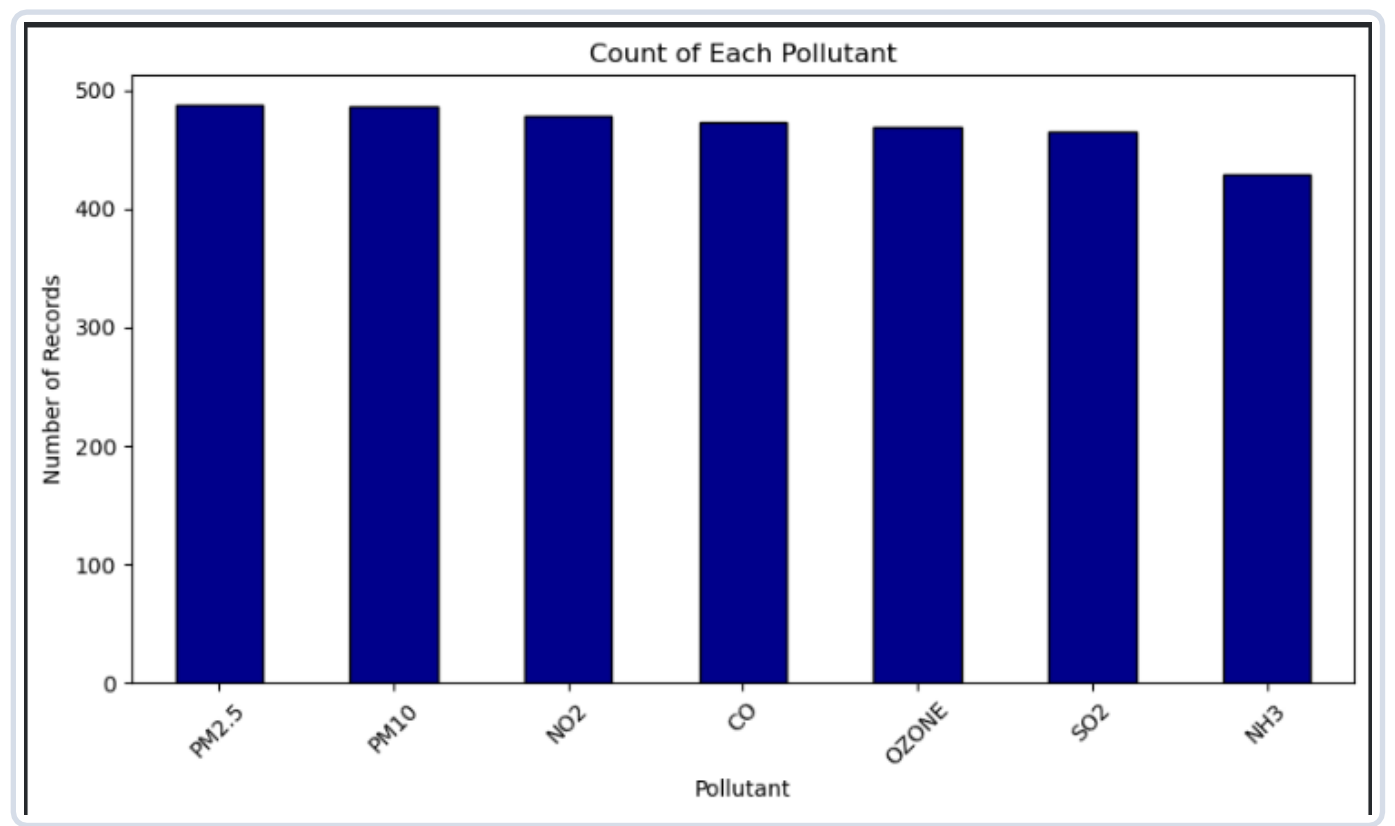
Further analysis showed that the **highest average pollutant level** was recorded for **OZONE** in **Angul, Odisha**, with an average value of **312**. This indicates that Ozone pollution was particularly severe in that location compared to other cities. Similarly, the **highest maximum pollutant value** was **500**, recorded for **PM2.5** in **Charkhi Dadri, Haryana**. PM2.5 consists of extremely fine particles that can enter deep into the lungs, making high concentrations a serious health concern.

In addition to pollution indicators, the weather dataset provides important environmental KPIs. The **average temperature** recorded across all weather stations is **32.35°C**, indicating generally warm conditions during the observation period. The **average humidity** is **38.67%**, showing moderate moisture levels in the atmosphere. The **average visibility** is **10.96 km**, suggesting that visibility was generally good in most locations. Finally, the **average wind speed** is **20.95 km/h**, which influences how pollutants spread and disperse through the air.

These baseline KPIs provide a clear overview of the environmental conditions and serve as the foundation for the detailed analysis, visualizations, and dashboard insights presented later in the project.

Metric	Location / Pollutant	Value
Highest pollutant average	Angul, Odisha - OZONE	312
Highest pollutant maximum	Charkhi Dadri, Haryana - PM2.5	500
Average temperature	Weather dataset	32.35 degrees C
Average humidity	Weather dataset	38.67%
Average visibility	Weather dataset	10.96 km
Average wind speed	Weather dataset	20.95 km/h

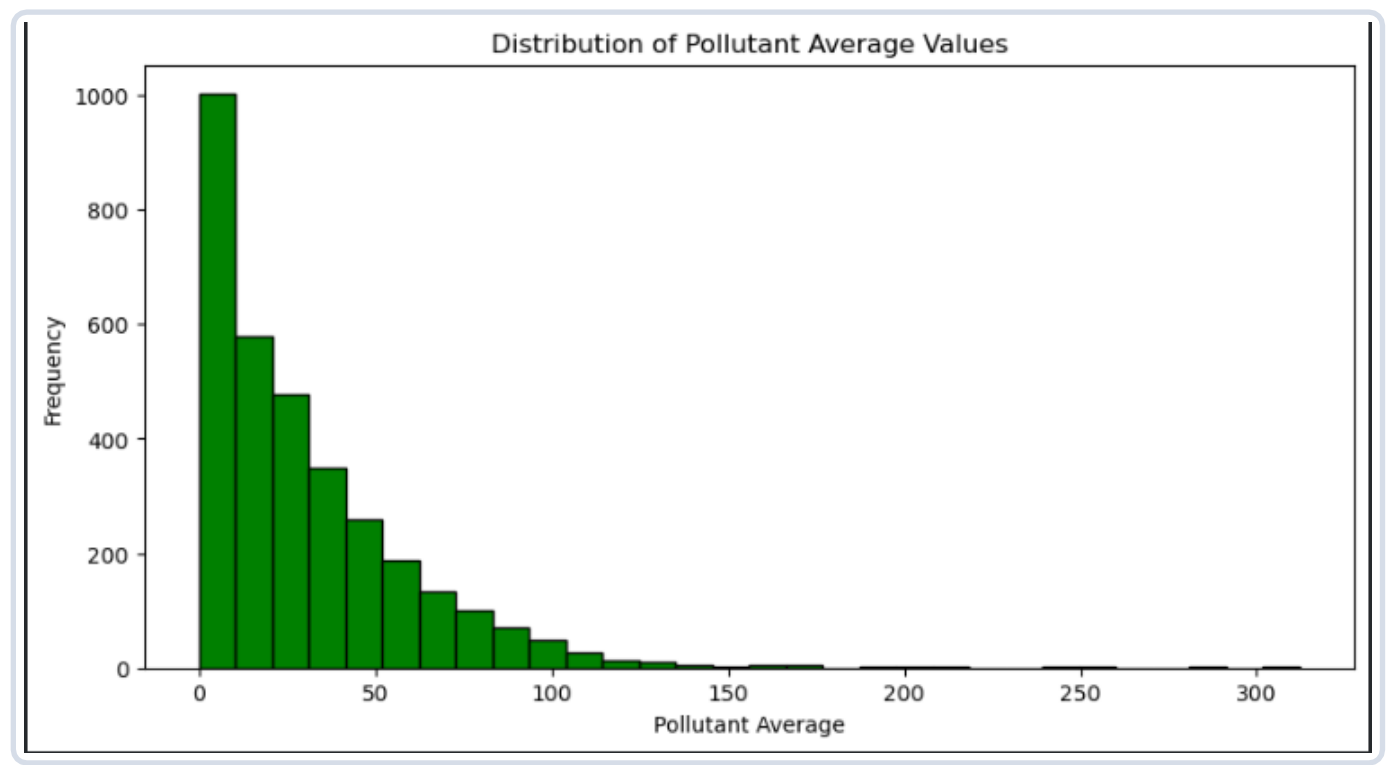
16.1 Count of Each Pollutant



INTERPRETATION

Part	Explanation
Method	The notebook counts rows for each pollutant id and displays the result as a bar chart.
What it shows	PM2.5 and PM10 have the highest number of records, while NH3 has the lowest record count.
Why it matters	The dataset is fairly balanced across pollutant types, so comparison between pollutants is meaningful.

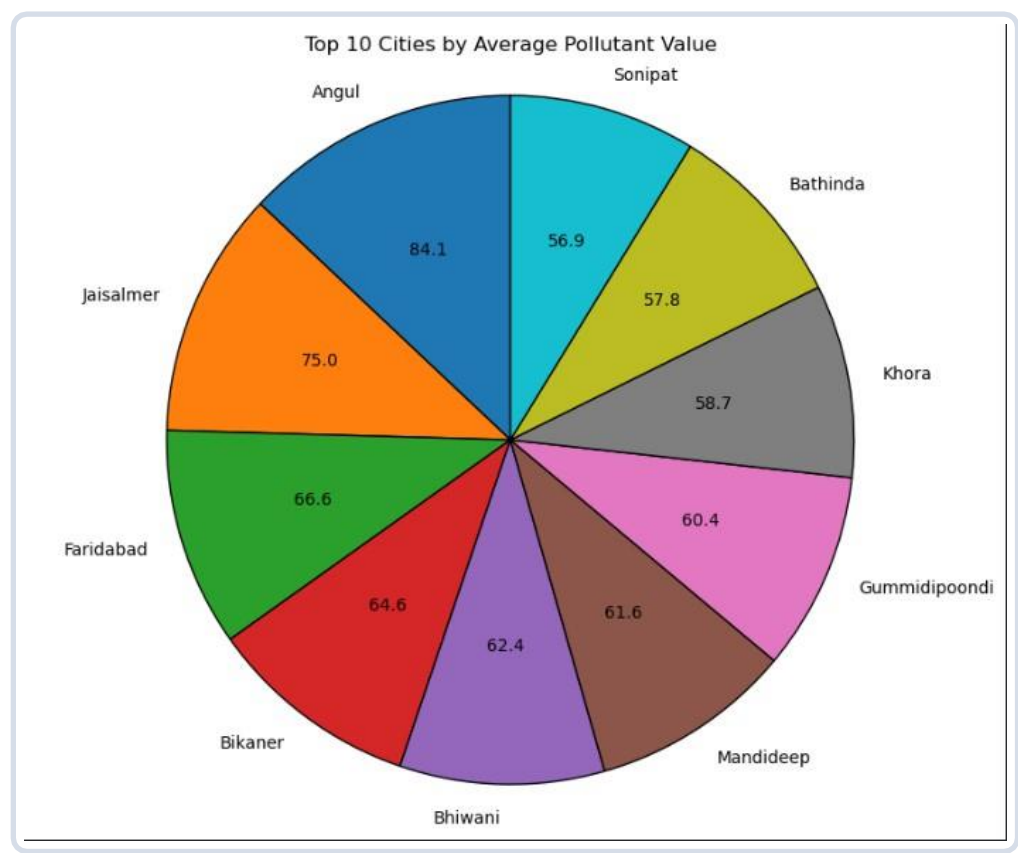
16.2 Distribution of Pollutant Average Values



INTERPRETATION

Part	Explanation
Method	The notebook plots pollutant average values in a histogram.
What it shows	Most pollutant average values are below 50, while a smaller number of records reach very high values.
Why it matters	The right-skewed shape means high pollution events are less common but important to monitor.

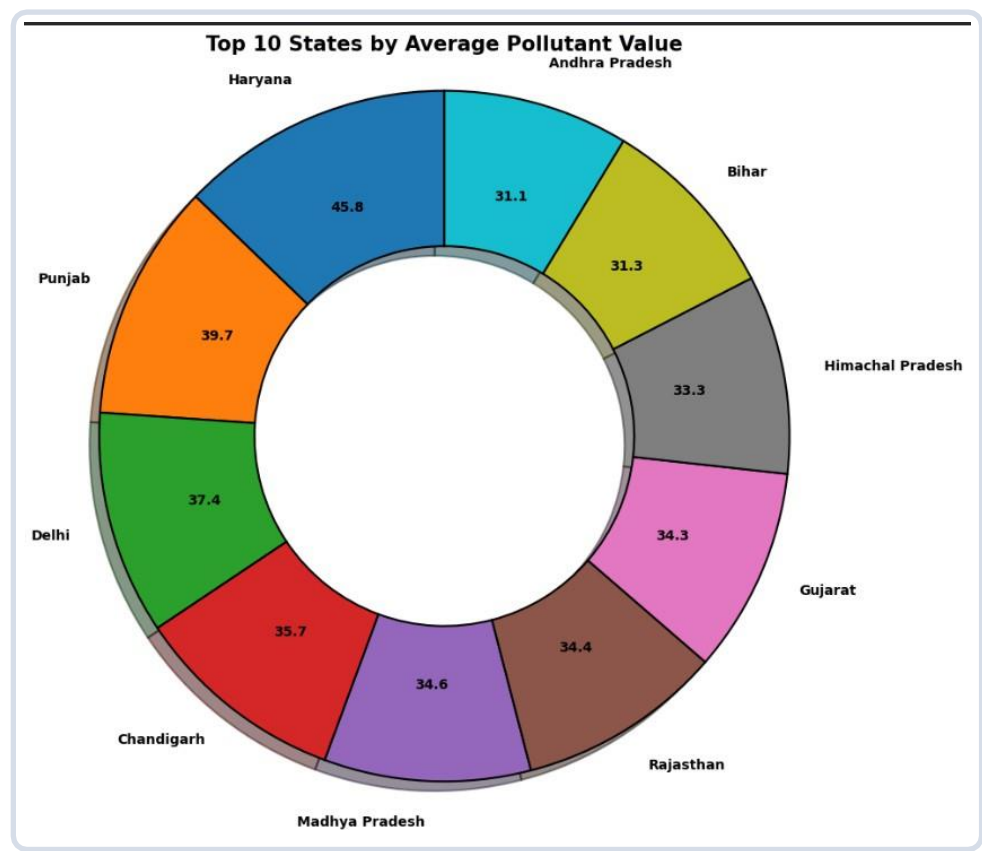
16.3 Top 10 Cities by Average Pollutant Value



INTERPRETATION

Part	Explanation
Method	The notebook groups data by city and calculates the mean pollutant average.
What it shows	Angul has the highest average pollutant value, followed by Jaisalmer, Faridabad, Bikaner, and Bhiwani.
Why it matters	City-level ranking helps identify where focused monitoring and pollution control may be needed first.

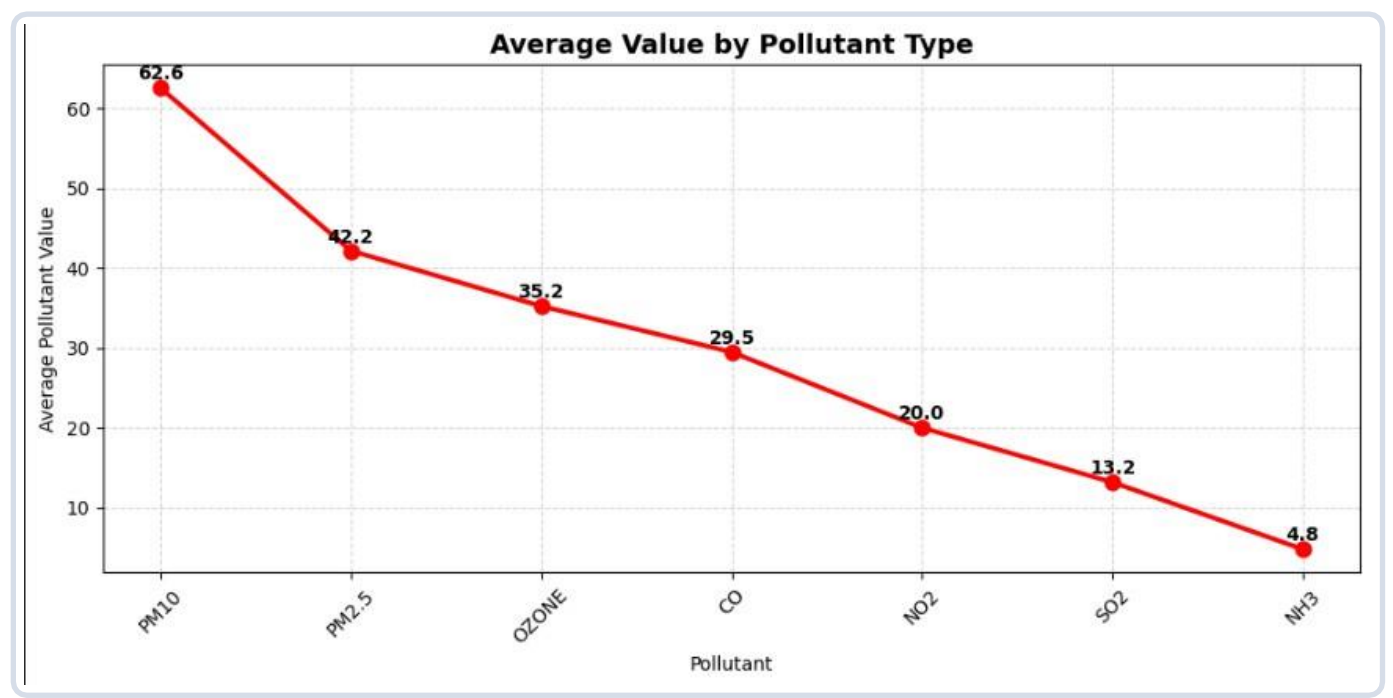
16.4 Top States by Average Pollutant Value



INTERPRETATION

Part	Explanation
Method	The notebook groups AQI records by state and calculates average pollutant value.
What it shows	Haryana is the highest state by average pollutant value, followed by Punjab, Delhi, and Chandigarh.
Why it matters	State-level comparison is useful for policy planning because pollution control is often managed regionally.

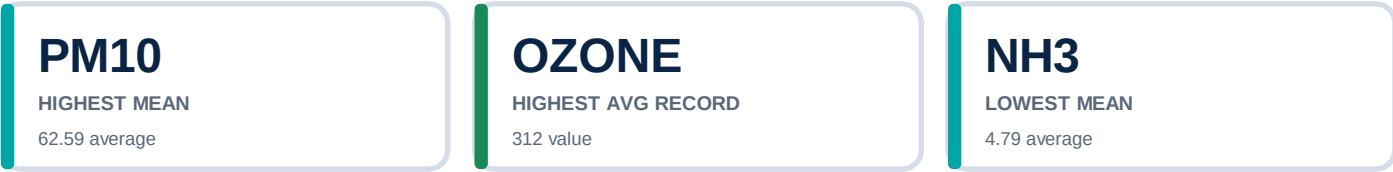
16.5 Average Value by Pollutant Type



INTERPRETATION

Part	Explanation
Method	The notebook groups records by pollutant id and calculates average pollutant value.
What it shows	PM10 has the highest average value at about 62.6, followed by PM2.5 at about 42.2. NH3 has the lowest average value.
Why it matters	The result shows that particulate matter, especially PM10 and PM2.5, is the most important pollutant group in this dataset.

16.6 Pollutant Summary Table



The **Pollutant Summary Table** provides a statistical overview of all the major pollutants present in the AQI dataset. It summarizes the number of observations, average pollution level, minimum value, and maximum value for each pollutant type. This table supports the pollutant-wise line chart shown in the dashboard by providing the exact numerical values behind the visualization.

The **Count** column shows how many AQI records are available for each pollutant. The **Mean Avg** column represents the average value of the **pollutant_avg** field and indicates the overall pollution level for each pollutant across all locations. The **Min Avg** column shows the lowest recorded average pollutant value, while the **Max Avg** column represents the highest recorded average value for that pollutant.

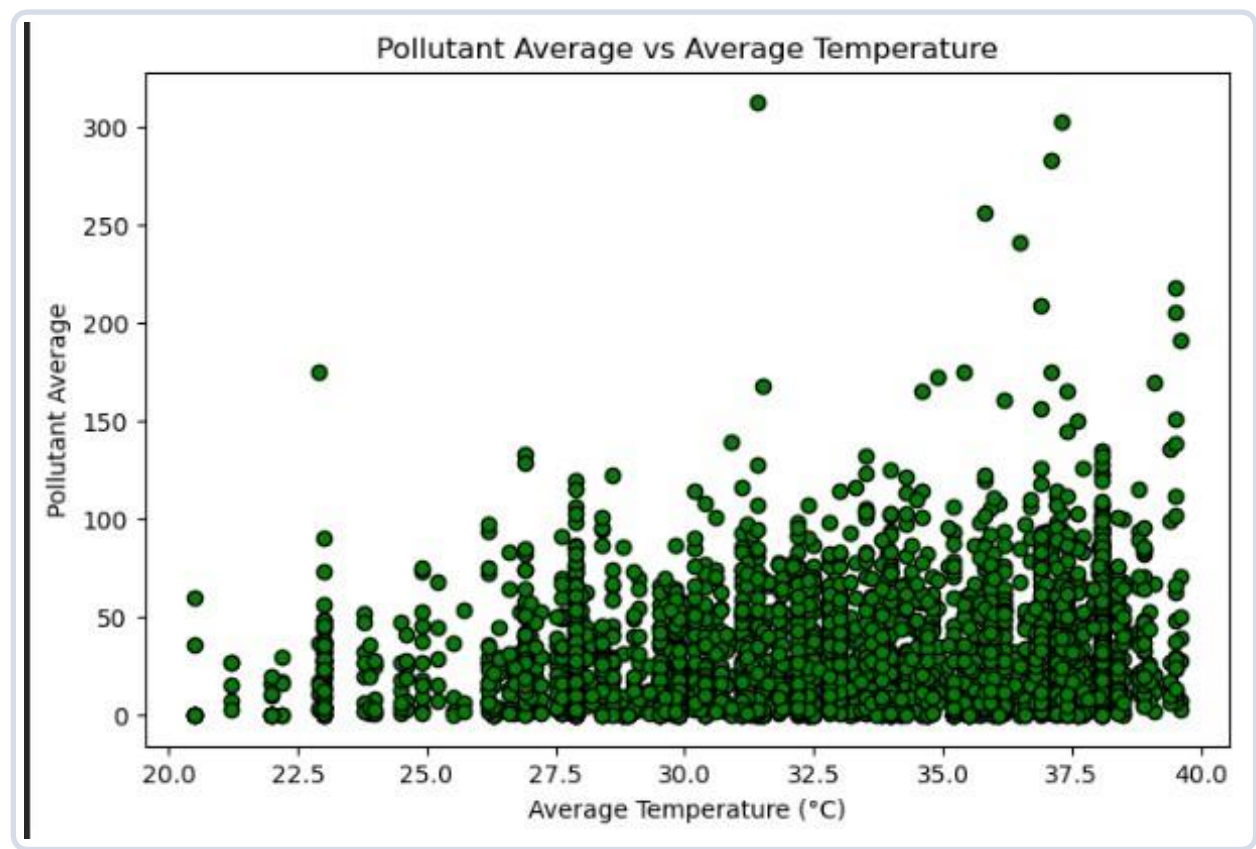
The analysis shows that **PM10** has the highest average pollution level with a **Mean Avg of 62.59**, making it the most dominant pollutant in the dataset. **PM2.5** ranks second with an average value of **42.19**, indicating that fine particulate matter is also a major contributor to air pollution. **OZONE** ranks third with an average value of **35.25**, followed by **CO (Carbon Monoxide)** with **29.50** and **NO₂ (Nitrogen Dioxide)** with **20.01**.

Among all the pollutants, **SO₂ (Sulphur Dioxide)** has a comparatively lower average value of **13.19**, while **NH₃ (Ammonia)** records the lowest average pollution level at **4.79**, making it the least dominant pollutant in this project.

The minimum average value for every pollutant is **0**, indicating that some monitoring stations recorded negligible or no concentration of that pollutant during the observation period. However, the maximum values vary significantly, showing that pollution levels can become very high in certain locations. For example, **OZONE** reaches the highest maximum average value of **312**, while **PM10** reaches **283** and **PM2.5** reaches **256**, highlighting locations where these pollutants were present in much higher concentrations.

Pollutant	Count	Mean Avg	Min Avg	Max Avg
PM10	486	62.59	0	283
PM2.5	488	42.19	0	256
OZONE	469	35.25	0	312
CO	473	29.50	0	123
NO2	478	20.01	0	122
SO2	465	13.19	0	92
NH3	429	4.79	0	23

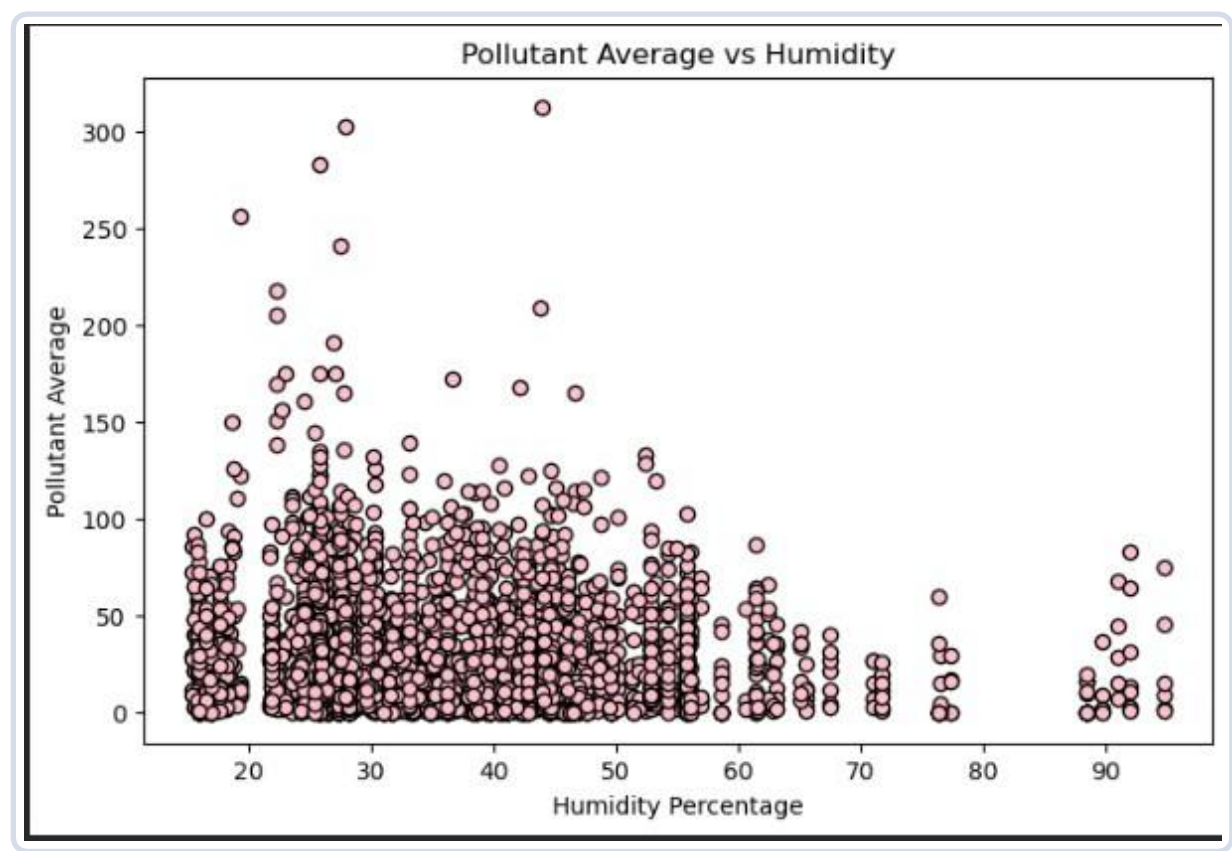
16.7 Pollutant Average vs Temperature



INTERPRETATION

Part	Explanation
Method	The notebook joins weather information and plots pollutant average against average temperature.
What it shows	Most records are between about 28 and 38 degrees C, and most pollutant values are below 100.
Why it matters	Temperature alone does not explain pollution. Traffic, dust, industry, rainfall, and wind also matter.

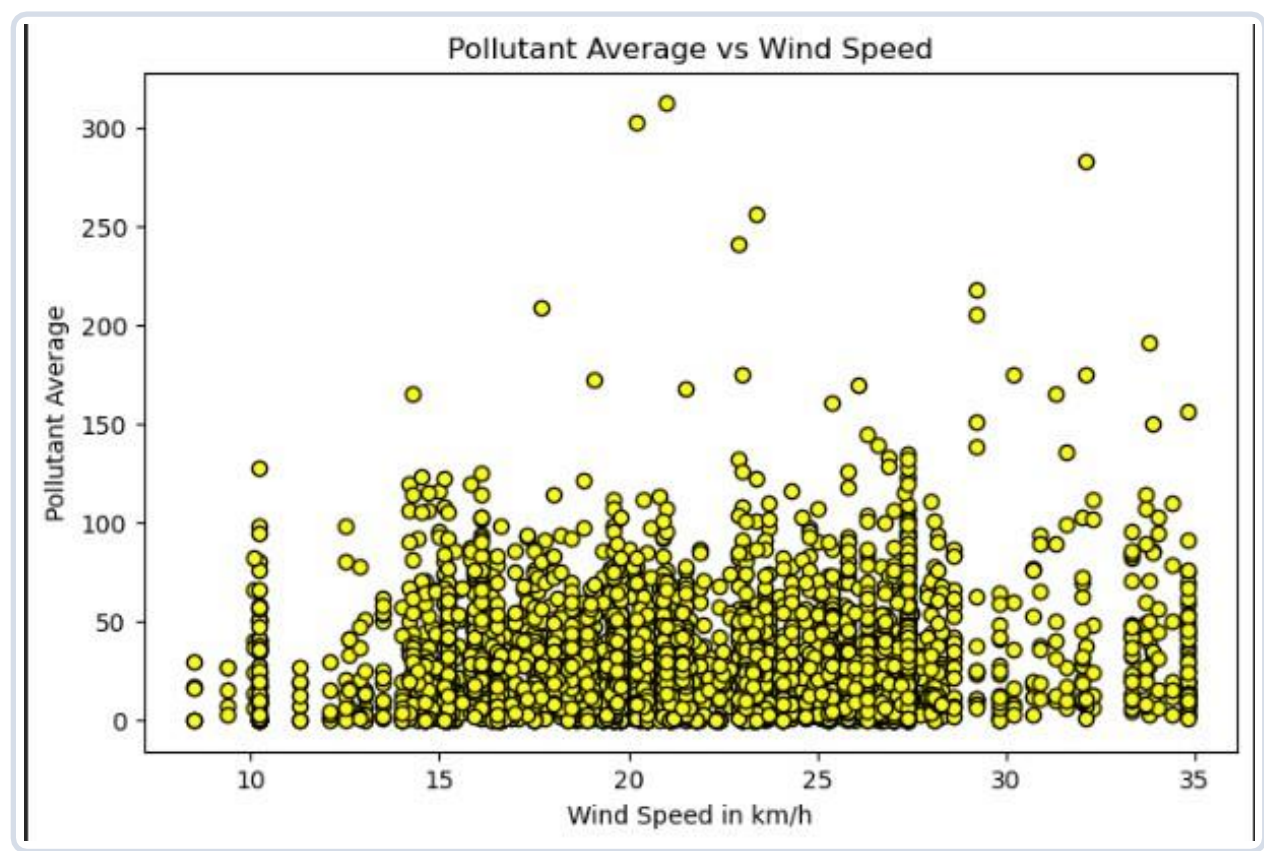
16.8 Pollutant Average vs Humidity



INTERPRETATION

Part	Explanation
Method	The notebook plots pollutant average against humidity percentage.
What it shows	Pollution values are spread across many humidity levels. High and low pollutant values appear in different humidity ranges.
Why it matters	Humidity has a weak relationship with pollution in this dataset, so it should be interpreted with other factors.

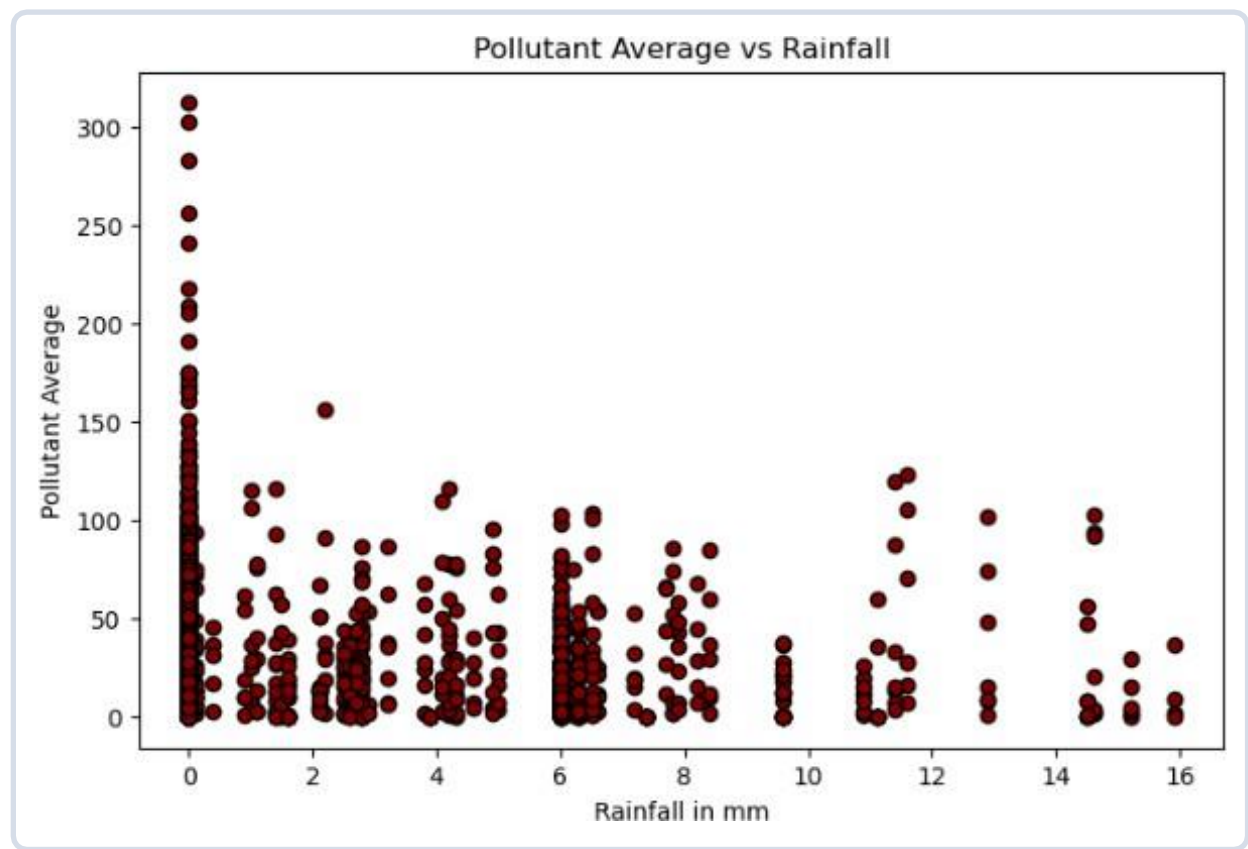
16.9 Pollutant Average vs Wind Speed



INTERPRETATION

Part	Explanation
Method	The notebook plots pollutant average against wind speed in km/h.
What it shows	Wind speed values mostly sit between about 10 and 35 km/h. Pollution values still vary widely within that range.
Why it matters	Wind can spread pollutants, but local emissions and weather conditions also influence the final pollutant value.

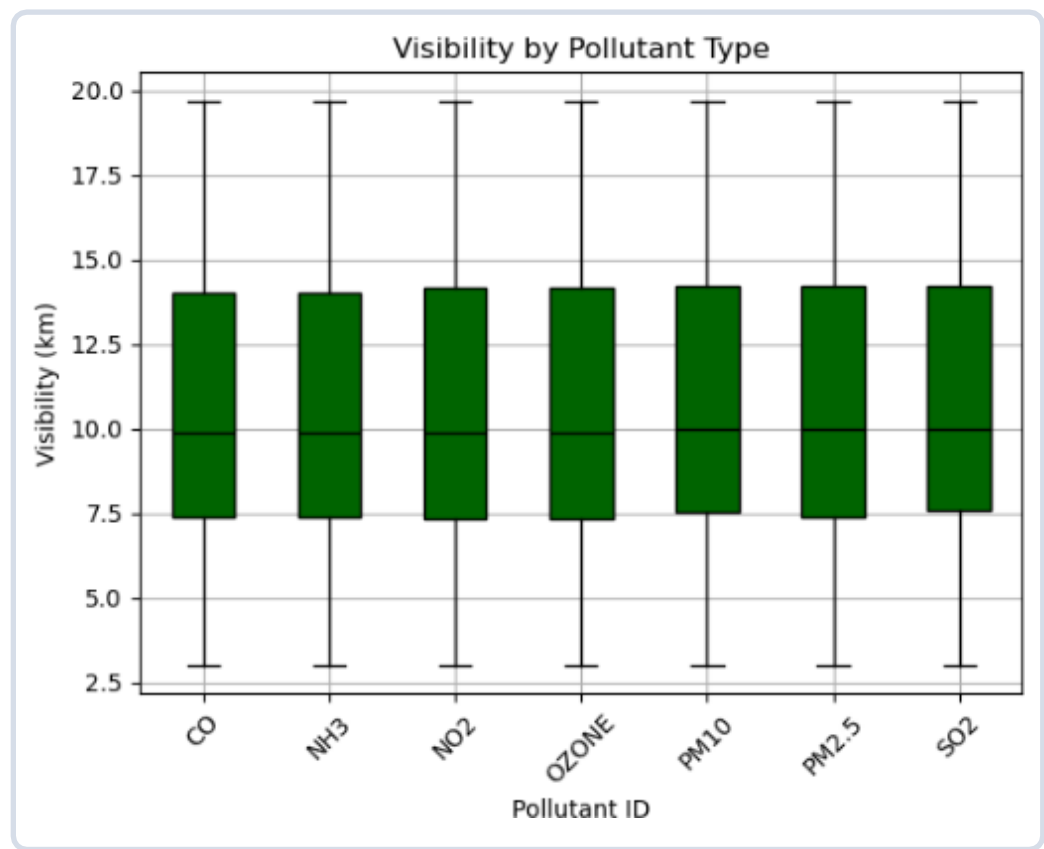
16.10 Pollutant Average vs Rainfall



INTERPRETATION

Part	Explanation
Method	The notebook compares pollutant average with rainfall in millimeters.
What it shows	Many records have zero or very low rainfall. Higher rainfall records generally do not dominate the highest pollutant values.
Why it matters	Rainfall may reduce dust and particles, but this single-date dataset should not be treated as proof of a strong rainfall effect.

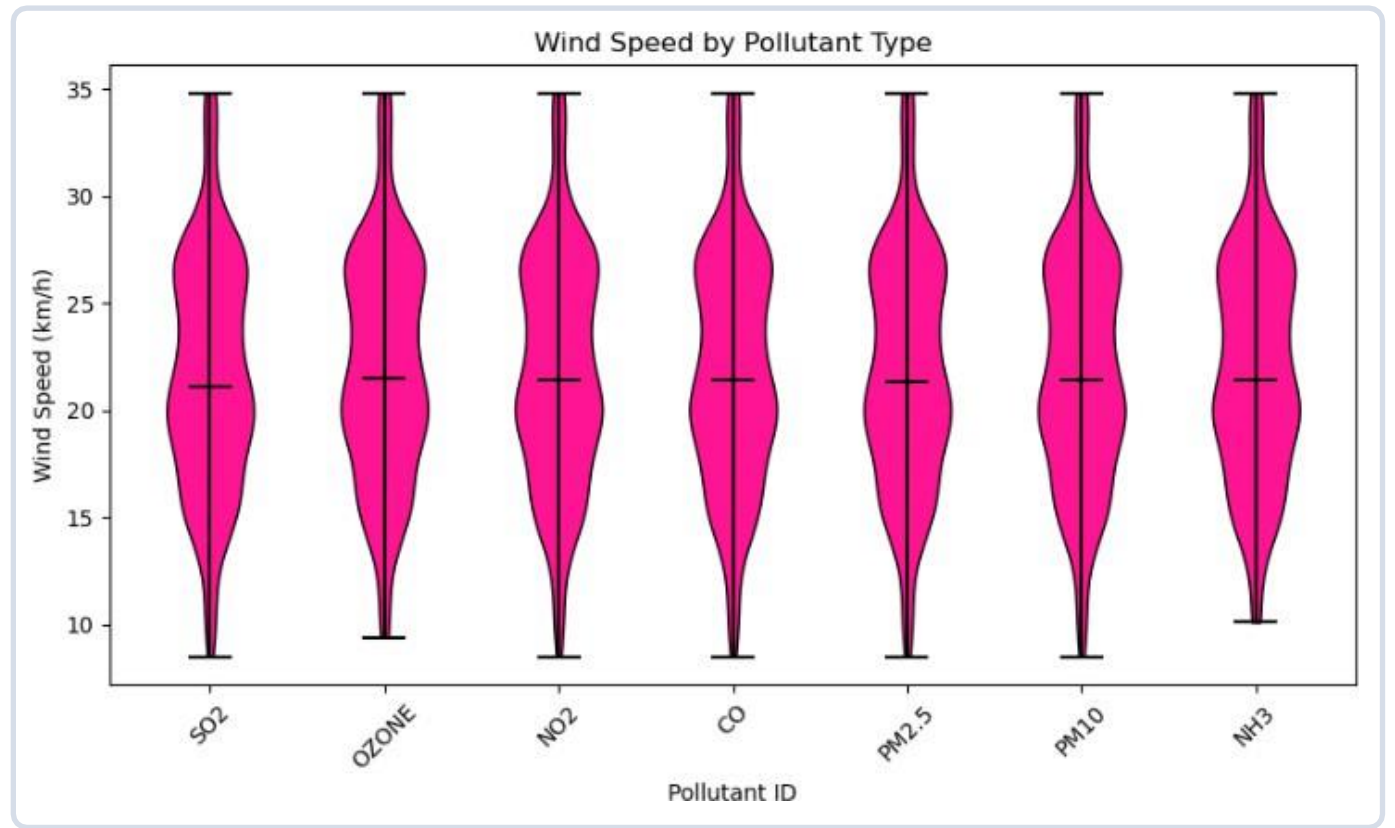
16.11 Visibility by Pollutant Type



INTERPRETATION

Part	Explanation
Method	The notebook uses a box plot to compare visibility across pollutant types.
What it shows	The visibility distribution is very similar for CO, NH3, NO2, OZONE, PM10, PM2.5, and SO2.
Why it matters	In this dataset, no single pollutant type clearly shows a much lower visibility pattern than the others.

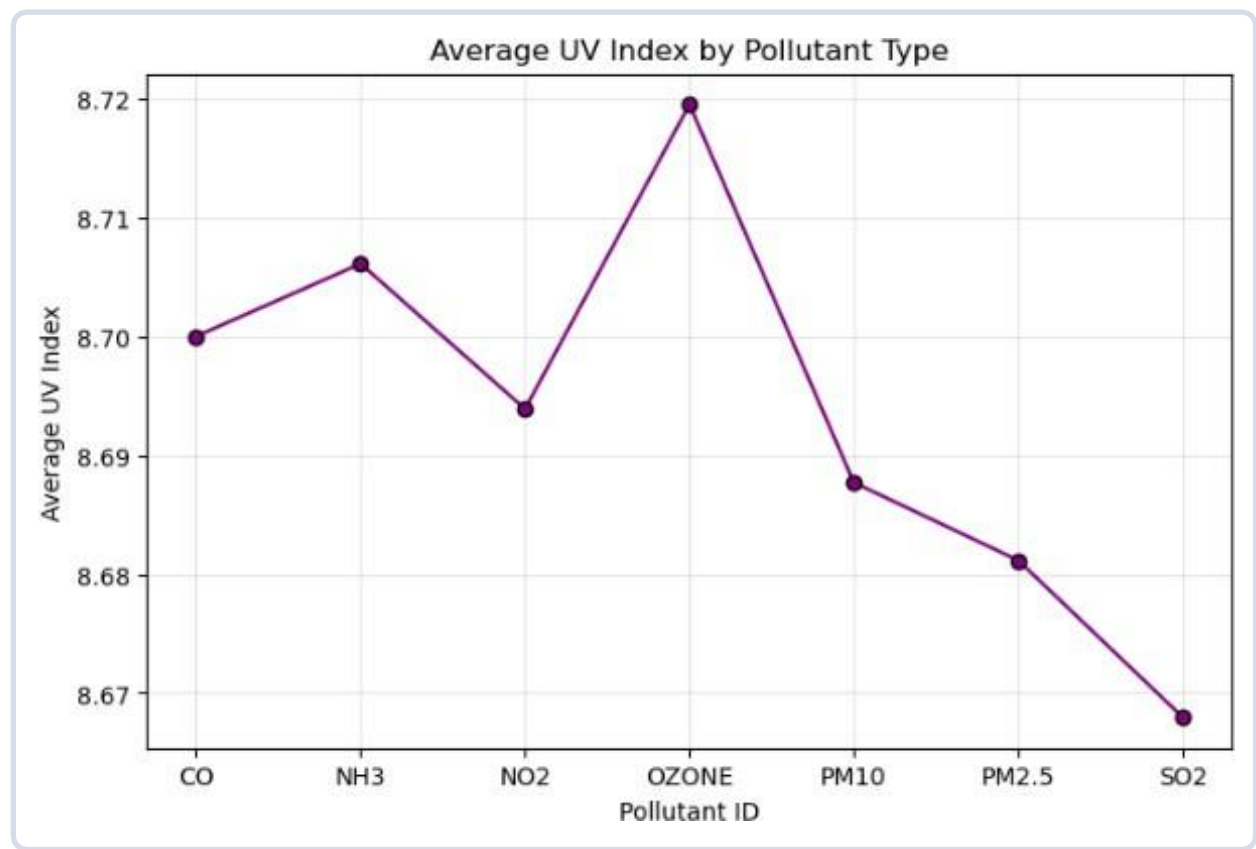
16.12 Wind Speed by Pollutant Type



INTERPRETATION

Part	Explanation
Method	The notebook uses a violin plot to compare wind speed distribution by pollutant id.
What it shows	Wind speeds are quite similar across all pollutant types, mostly centered around the same middle range.
Why it matters	This suggests wind speed is a common weather condition across pollutant records rather than a pollutant-specific feature.

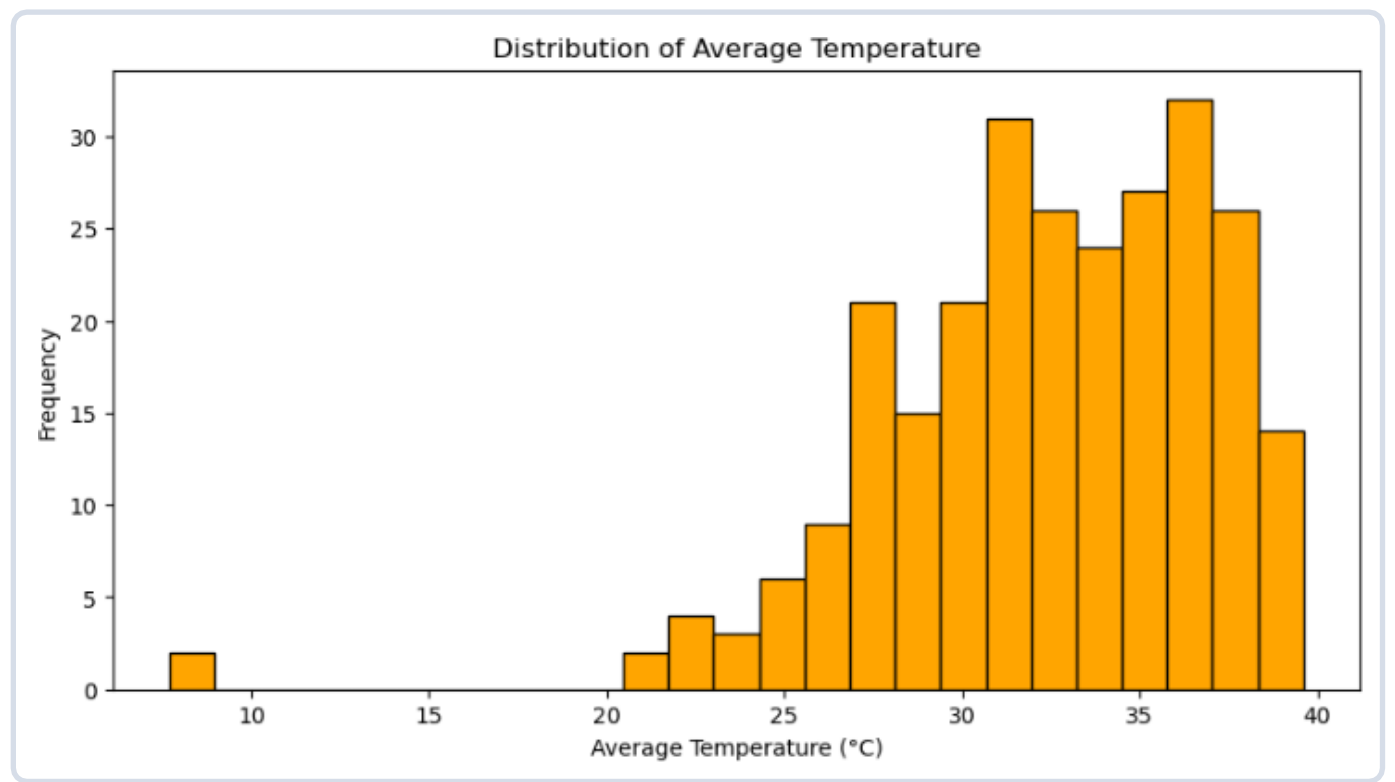
16.13 Average UV Index by Pollutant Type



INTERPRETATION

Part	Explanation
Method	The notebook calculates the average UV index for each pollutant type and plots the comparison.
What it shows	The UV index values are close to each other across pollutant types, with OZONE slightly higher in the chart.
Why it matters	UV index may support ozone-related interpretation, but the differences are small and should be treated carefully.

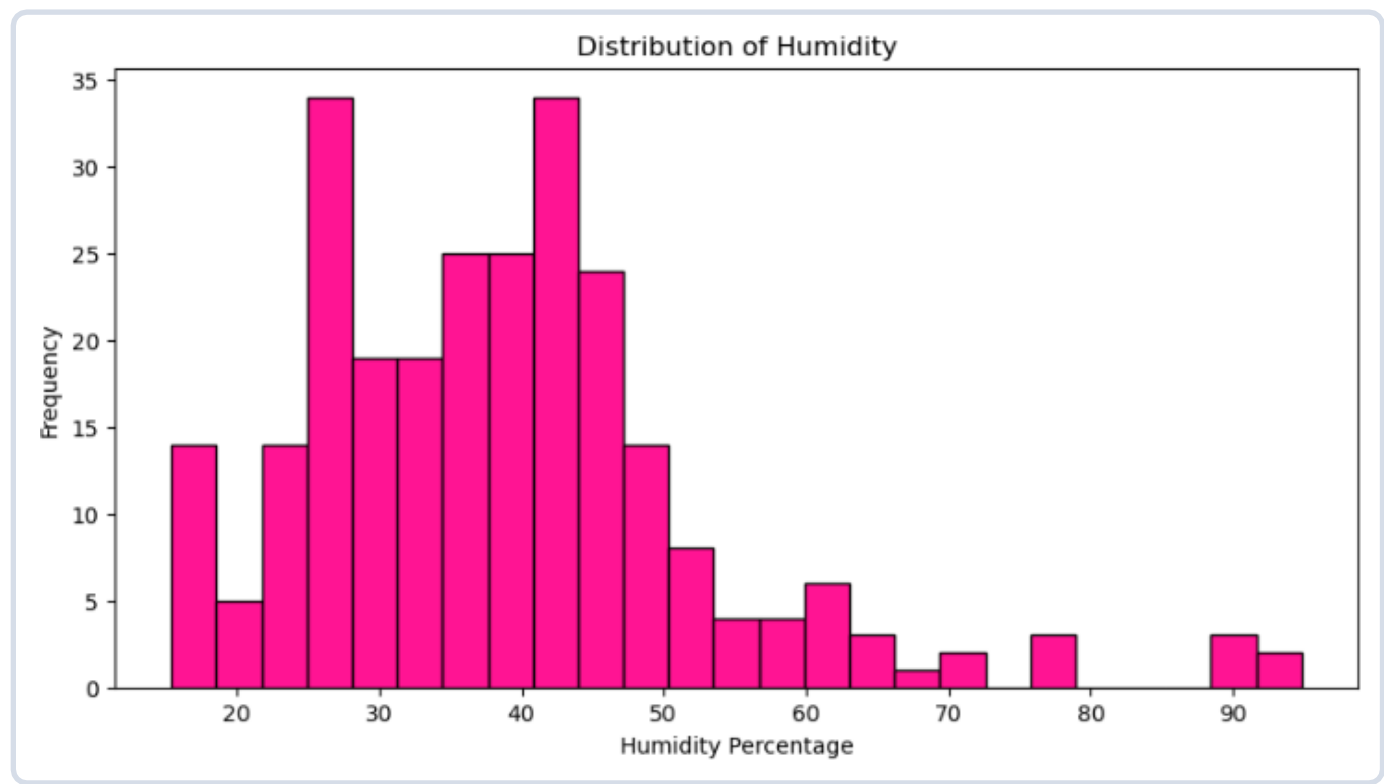
16.14 Distribution of Average Temperature



INTERPRETATION

Part	Explanation
Method	The notebook plots the distribution of average temperature values in the weather dataset.
What it shows	Most cities have average temperature values roughly between the high 20s and high 30s degrees C.
Why it matters	This helps explain the climate context of the pollution readings for 31-05-2026.

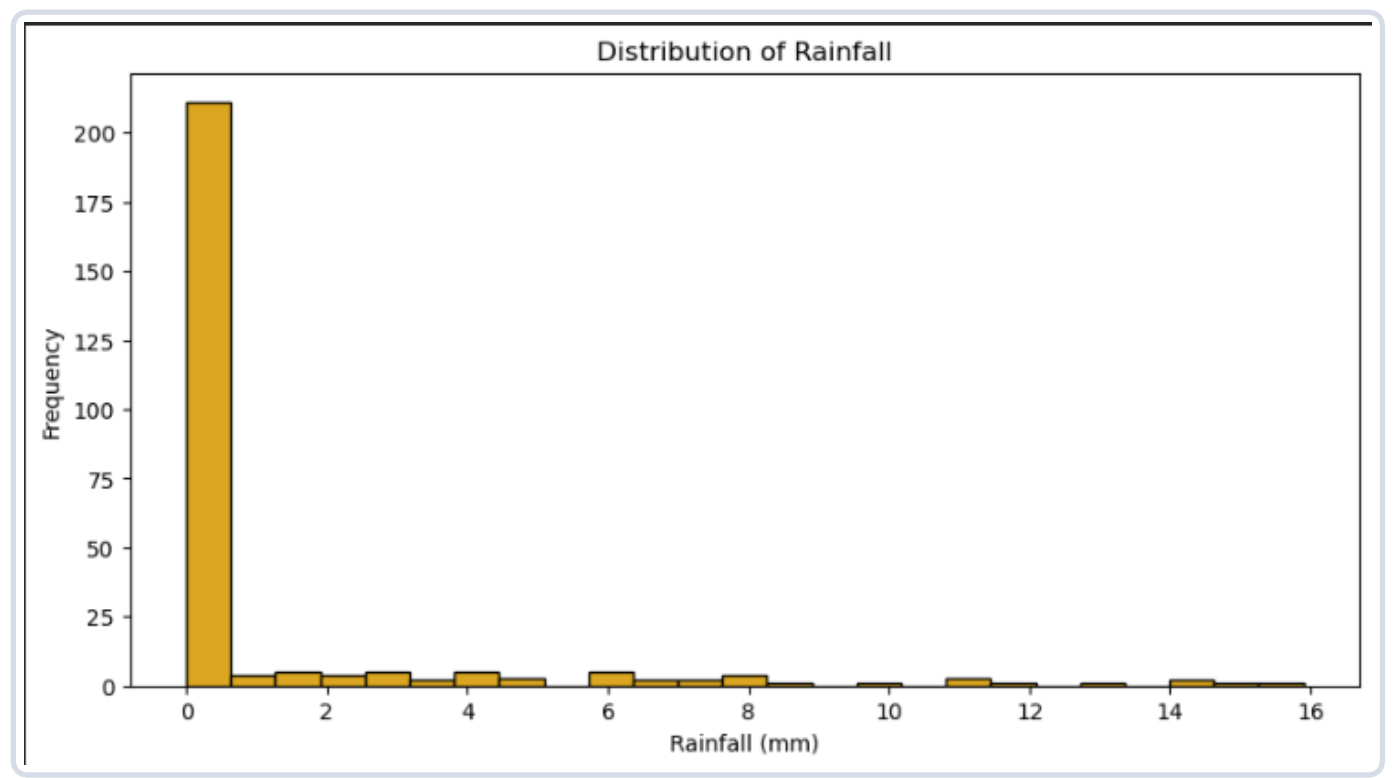
16.15 Distribution of Humidity



INTERPRETATION

Part	Explanation
Method	The notebook plots humidity percentage as a histogram.
What it shows	Most humidity readings fall between about 25% and 50%, with some high-humidity locations also present.
Why it matters	Humidity is useful context because it can change how particles stay in the air, but it is not a standalone answer.

16.16 Distribution of Rainfall



INTERPRETATION

Part	Explanation
Method	The notebook plots rainfall values in the weather dataset.
What it shows	Most records show zero rainfall, while a small group of cities show rainfall up to about 16 mm.
Why it matters	This explains why rainfall has limited influence in many charts: most locations were dry on the data date.

16.17 Weather Correlation Summary

The **Weather Correlation Summary** explains how different weather conditions are related to the **average pollutant value** in the AQI dataset. Correlation analysis is a statistical method used to measure whether two numerical variables change together. It helps identify whether changes in weather conditions are associated with changes in pollution levels.

A **correlation value** ranges from **-1 to +1**:

- A value **close to +1** indicates a **strong positive relationship**, meaning both variables increase together.
- A value **close to -1** indicates a **strong negative relationship**, meaning when one variable increases, the other decreases.
- A value **close to 0** indicates **little or no relationship**, meaning the variables are mostly independent of each other.

In this project, the correlation analysis shows that **none of the weather factors have a strong relationship with the average pollutant value**. All correlation values are relatively close to zero, indicating that weather conditions alone do not have a significant impact on pollution levels in this dataset. Other factors such as traffic density, industrial activities, construction work, and local environmental conditions may have a greater influence on air pollution.

Among all the weather variables, **average temperature** has the **highest positive correlation (0.170)** with pollutant levels. Although this is the strongest relationship in the dataset, it is still considered **weak**, suggesting that slightly higher temperatures may be associated with a small increase in pollution levels.

Wind speed (0.103) and **UV index (0.069)** also show **very weak positive relationships**, indicating that these factors have only a minor association with pollutant concentrations.

On the other hand, **humidity (-0.135)** and **cloud cover (-0.115)** have the strongest negative correlations, although these are still weak. This suggests that as humidity or cloud cover increases, pollutant levels may decrease slightly, but the relationship is not strong enough to draw firm conclusions.

Similarly, **rainfall (-0.081)** and **visibility (-0.068)** show very weak negative relationships, while **atmospheric pressure (-0.031)** has almost no correlation with pollutant levels.

Overall, the correlation analysis indicates that **weather variables have only a limited influence on air pollution in this dataset**, and pollution levels are likely affected more by human activities and emission sources than by weather conditions alone.

Weather Correlation Summary

Weather Factor	Correlation with Pollutant Avg	Simple Reading
temperature avg in C	0.170	Weak positive relationship
wind speed in km/h	0.103	Very weak positive relationship
uv index	0.069	Very weak positive relationship
pressure in h/Pa	-0.031	Almost no relationship
visibility in km	-0.068	Very weak negative relationship
rainfall in mm	-0.081	Very weak negative relationship
cloud cover Pct	-0.115	Weak negative relationship
humidity Pct	-0.135	Weak negative relationship

16.18 Python EDA Outcome Summary

The **Exploratory Data Analysis (EDA)** performed in Python provided a comprehensive understanding of the Air Quality Index (AQI) and Weather datasets. Using statistical summaries and a variety of visualizations, the analysis helped identify pollution patterns, compare cities and states, examine pollutant distributions, and study the relationship between weather conditions and air pollution. Since all datasets were cleaned and validated before analysis, the insights obtained are accurate, reliable, and suitable for decision-making.

The EDA confirms that the pollutant records are **well distributed across the different pollutant categories**, allowing meaningful comparisons between pollutants. Among all the pollutant types, **PM10** has the highest average pollutant value, followed by **PM2.5**. These particulate matter pollutants are the major contributors to poor air quality in the dataset and are known to have serious health impacts because they can remain suspended in the air and be inhaled into the respiratory system.

The city-wise analysis identifies **Angul, Jaisalmer, Faridabad, Bikaner, and Bhiwani** as some of the most polluted cities based on their average pollutant values. These cities consistently appear among the highest-ranking locations in both the statistical analysis and the dashboard visualizations, indicating that they should be considered priority areas for pollution monitoring and environmental improvement initiatives.

The state-wise analysis shows that **Haryana** has the highest **average pollutant value**, indicating higher overall pollution levels compared to other states. However, when the Power BI dashboard calculates pollution using the **sum of pollutant_max values**, **Rajasthan** ranks first. This difference occurs because average values and summed maximum values represent different methods of measuring pollution intensity, and both provide useful but different perspectives on the data.

The weather analysis indicates that **temperature, humidity, rainfall, wind speed, visibility, and UV index** have only **weak correlations** with the average pollutant values. Although temperature shows the strongest positive relationship among the weather variables, its influence is still relatively small. These findings suggest that weather conditions alone cannot explain pollution levels. Instead, air quality is influenced by a combination of factors such as **vehicle emissions, industrial activities, construction dust, population density, geographical conditions, and atmospheric movement**.

An important limitation of the analysis is that the dataset represents observations from **31-05-2026** only. Therefore, the project performs a **cross-sectional comparison** of different cities and states on a single day rather than studying pollution changes over time. As a result, the findings are highly useful for comparing locations and identifying pollution hotspots but should **not** be interpreted as long-term pollution trends or forecasting results.

Key Findings

- **PM10** records the highest average pollutant value, making it the most dominant pollutant in the dataset.
- **PM2.5** is the second most significant pollutant and is another major contributor to poor air quality.
- **Angul, Jaisalmer, Faridabad, Bikaner, and Bhiwani** are identified as the cities with the highest pollution levels.
- **Haryana** has the highest average pollutant value among all states.
- **Rajasthan** ranks highest in the Power BI dashboard when pollution is measured using the **sum of pollutant_max values**.
- Weather variables show **only weak relationships** with pollutant levels, indicating that pollution depends on multiple environmental and human-related factors rather than weather alone.
- The analysis is based on **one day's data (31-05-2026)**, making it ideal for **city and state comparisons**, but **not suitable for long-term trend analysis or future forecasting**.
- The insights generated from the Python EDA provide a strong analytical foundation for the Power BI dashboards and help support informed environmental planning and pollution-control decisions.

17. Power BI Dashboard Design

The **Exploratory Data Analysis (EDA)** performed in Python provided a clear understanding of the air quality and weather datasets. By using statistical summaries and different visualizations, the analysis helped identify pollution patterns, compare cities and states, and study the relationship between weather conditions and pollutant levels. Since the datasets were already cleaned and validated, the results obtained from the analysis are reliable and suitable for building dashboards and drawing meaningful conclusions.

The analysis shows that the pollutant records are fairly balanced across the different pollutant categories, allowing a fair comparison between them. Among all pollutants, **PM10** has the highest average pollutant value, followed by **PM2.5**. These particulate matter pollutants are the major contributors to poor air quality and require greater attention because they can remain suspended in the air and have serious effects on human health.

The city-wise analysis highlights **Angul, Jaisalmer, Faridabad, Bikaner, and Bhiwani** as some of the locations with the highest pollution levels in the dataset. These cities recorded consistently high pollutant values compared to many other locations and therefore require closer monitoring and pollution control measures.

The state-wise analysis shows that **Haryana** has the highest **average pollutant value**, indicating that, on average, pollution levels are higher across its monitoring stations than in other states. However, when the dashboard summarizes pollution using the **sum of pollutant maximum values**, **Rajasthan** records the highest total. This difference occurs because average values and total maximum values measure pollution in different ways and therefore highlight different aspects of the data.

The weather analysis indicates that **temperature, humidity, rainfall, wind speed, visibility, and UV index** have only **weak relationships** with pollutant levels. None of these weather factors individually explain pollution levels in a significant way. This is expected because air quality is influenced by many factors working together, including **vehicle emissions, industrial activities, construction dust, population density, geographical conditions, and wind movement**, rather than by weather conditions alone.

It is also important to note that the analysis is based on data collected for **a single date (31-05-2026)**. Therefore, the project is best suited for comparing pollution levels across different cities, states, and pollutant types on that specific day. Since historical data is not available, the project is **not intended for long-term trend analysis or future pollution forecasting**.

Key Findings

- **PM10** has the highest average pollutant value, making it the most dominant pollutant in the dataset.
- **PM2.5** is the second-highest pollutant and is also a major contributor to poor air quality.
- **Angul, Jaisalmer, Faridabad, Bikaner, and Bhiwani** are among the cities with the highest pollution levels.
- **Haryana** records the highest average pollutant value among all states.
- **Rajasthan** ranks highest when pollution is measured using the **sum of pollutant maximum values** in the dashboard.
- Weather factors show only **weak correlations** with pollution levels, indicating that pollution depends on multiple environmental and human-related factors.
- The analysis is based on **one day's data**, making it ideal for **cross-sectional comparison** but not for predicting long-term pollution trends.

Overall, the Python EDA successfully transformed the raw datasets into meaningful insights, helping identify pollution hotspots, compare environmental conditions across locations, and provide a strong analytical foundation for the Power BI dashboards and final project report.

This dashboard page gives a high-level view of air pollution. It shows average pollutant value, maximum pollutant value, total monitoring stations, total cities, average pollution by state and city, pollutant ranking, and total pollutant values by state.



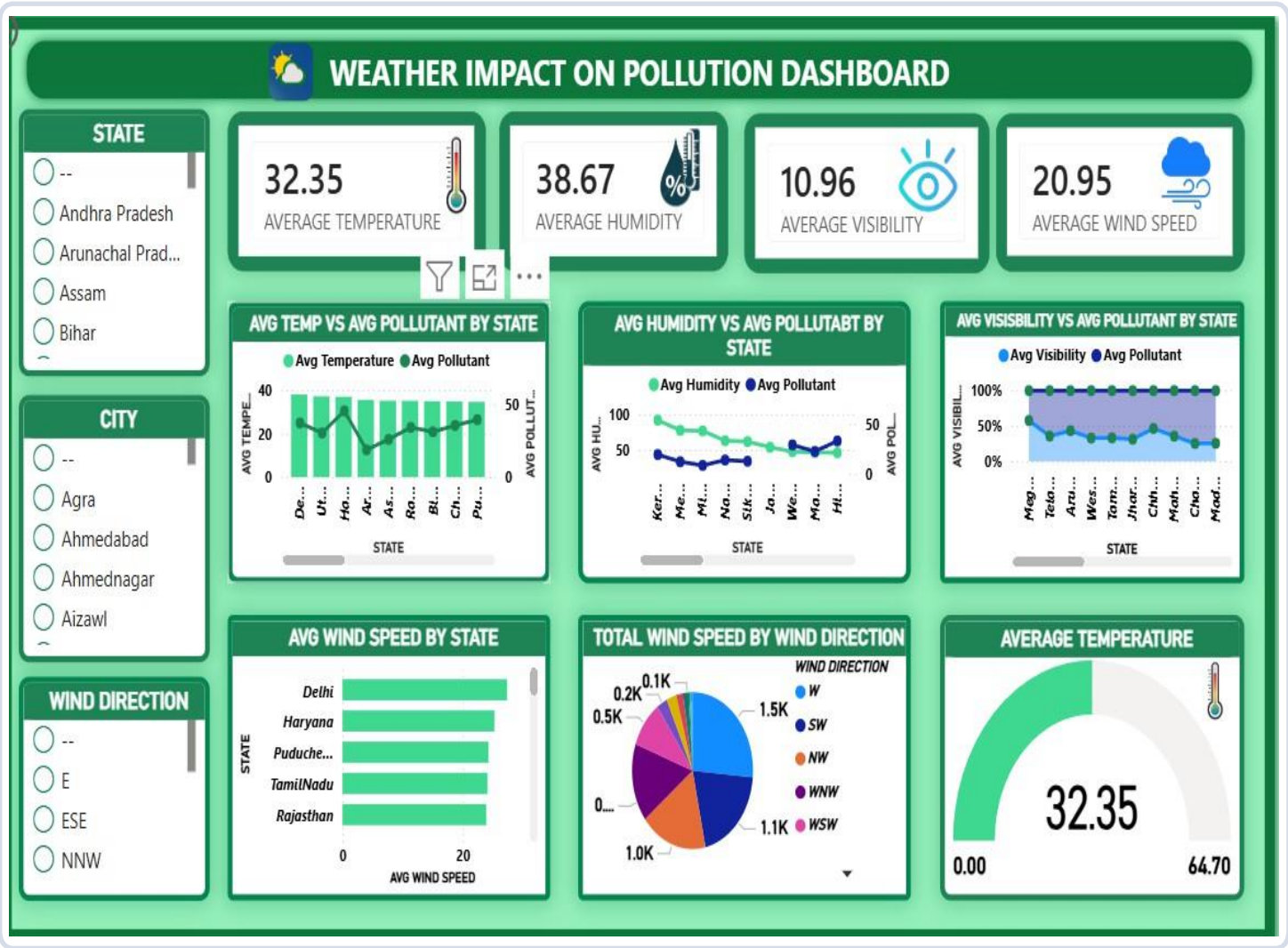
17.2 AQI Dashboard Interpretation

The first dashboard is useful for quickly finding high-pollution areas and dominant pollutants. It also separates average, minimum, maximum, and total views so the user can understand pollution from different angles.

Visual	Displayed Result	Simple Interpretation
Average Pollutant KPI	30.18	Overall mean pollutant average across all AQI records
Max Pollutant KPI	500	Maximum value from pollutant_max field
Monitoring Stations	494	Number of unique station names in AQI data
Average by State	Haryana highest at about 46	Haryana has the highest average pollutant value in the dashboard view
Average by City	Angul highest at about 84	Angul is the top city by average pollutant value
Average by Pollutant ID	PM10 highest, NH3 lowest	Particulate matter is the main pollutant concern
Total by State	Rajasthan about 30K	This visual uses sum of pollutant_max, not the average field
Minimum by City	Delhi highest	This visual uses sum of pollutant_min by city

17.3 Weather Impact Dashboard

This dashboard page focuses on weather and pollution together. It shows average temperature, average humidity, average visibility, average wind speed, and charts that compare these weather factors with pollutant values.



17.4 Weather Dashboard Interpretation

The second dashboard helps users understand the environmental context behind the AQI values. It does not prove that one weather factor directly causes pollution, but it helps compare weather and pollution side by side.

Visual	Displayed Result	Simple Interpretation
Average Temperature	32.35 degrees C	Overall mean temperature in the weather dataset
Average Humidity	38.67%	Overall mean humidity
Average Visibility	10.96 km	Average visibility across weather records
Average Wind Speed	20.95 km/h	Average wind speed across cities
Temp vs Pollutant	Mixed state pattern	Higher temperature does not always mean higher pollution
Humidity vs Pollutant	Weak relationship	Humidity alone is not enough to explain pollution
Wind Direction	West has highest total wind speed	Wind movement is dominated by a few directions
Gauge	32.35 on 0 to 64.70 scale	Shows average temperature as a quick dashboard summary

18. PPT Report Summary

The project PPT communicates the full story in presentation format. It starts with project overview and AQI meaning, then explains datasets, dataset relationships, workflow, business objectives, EDA insights, dashboard screenshots, recommendations, and conclusion.

The PPT is helpful for presenting the project verbally, while this PDF report gives a more detailed written explanation. Both outputs support the same project story.

PPT Section	Content Covered
Project Overview	Air pollution problem and purpose of AQI analysis
What is AQI	Simple definition and AQI categories
Datasets Available	Location_Master, AQI_Data, and Weather_Data roles
Project Workflow	Excel, PostgreSQL, Python, Power BI, PPT
EDA Insights	Pollutant counts, distributions, top cities/states, weather relationships
Dashboard Demonstration	AQI dashboard and weather impact dashboard
Recommendations	PM control, high-risk city focus, green cover, clean transport, public awareness

19. Business and Public Health Insights

The Air Quality Index (AQI) Analysis project provides valuable insights that can support **environmental monitoring, public health planning, and informed decision-making**. Instead of focusing on a single pollution value, the project combines multiple types of analysis, including **city-wise rankings, state-wise comparisons, pollutant analysis, weather observations, and interactive dashboard filtering**. This comprehensive approach helps users better understand pollution patterns and identify areas that require immediate attention.

The interactive dashboards allow users to easily compare pollution levels across different locations and pollutants. Decision-makers can use these insights to identify pollution hotspots, understand environmental conditions, and take appropriate actions to improve air quality.

Key Business and Public Health Insights

- **City-level analysis** helps identify cities with the highest pollution levels, allowing local authorities to focus monitoring efforts, improve pollution control measures, and plan targeted environmental actions.
- **Pollutant-level analysis** shows that **PM10** and **PM2.5** are the most significant pollutants in the dataset. Since these fine particles can seriously affect human health, reducing their concentration should be a priority for pollution control programs.
- **State-level comparisons** help government departments and environmental agencies identify regions with higher pollution levels. This enables better allocation of resources and supports state-wise planning for pollution reduction initiatives.
- **Weather analysis** provides additional context by showing how temperature, humidity, rainfall, wind speed, visibility, and other weather conditions vary across cities. Although the correlations are weak, these comparisons help explain why pollution levels may differ between locations on the same day.
- **Interactive Power BI dashboards** make complex data easy to understand. Users can apply filters, compare cities, states, and pollutant types, and explore the data visually without writing SQL queries or Python code.
- The project also supports **business and environmental decision-making** by providing reliable information that can assist urban planners, environmental consultants, researchers, and government organizations in identifying pollution trends and prioritizing improvement efforts.
- From a **public health perspective**, the findings highlight locations with poor air quality where awareness campaigns, health advisories, and preventive measures may be necessary. High pollution levels can increase the risk of respiratory illnesses, especially for children, older adults, and people with existing lung or heart conditions.
- The project demonstrates how data analytics can transform raw environmental data into meaningful insights that support evidence-based decisions, improve public awareness, and encourage sustainable environmental management.

Overall, the combination of **SQL, Python, and Power BI** provides a complete analytical solution that not only explains the current air quality conditions but also helps governments, businesses, and the public make better environmental and health-related decisions.

20. Key Findings

The analysis of the Air Quality Index (AQI) dataset revealed several important findings related to pollution levels across different cities, states, and pollutant types. These findings were consistently observed in both the **Python Exploratory Data Analysis (EDA)** and the **Power BI dashboards**, increasing confidence in the accuracy of the results. By combining statistical analysis with interactive visualizations, the project successfully identified pollution hotspots and highlighted the pollutants that contribute the most to poor air quality.

The analysis shows that **PM10** is the most dominant pollutant in the dataset, recording the highest average pollutant value among all pollutant categories. **PM2.5** also shows high pollution levels, confirming that particulate matter is the major air quality concern in the monitored locations.

The city-wise analysis identifies **Angul (Odisha)** as the most polluted city based on the **average pollutant value**, followed by **Jaisalmer, Faridabad, Bikaner, and Bhiwani**. These cities consistently recorded higher pollution levels than the other monitored locations, indicating that they may require greater environmental monitoring and pollution control measures.

The state-wise comparison shows that **Haryana** has the highest **average pollutant value**, making it the most polluted state when pollution is measured using average values. However, the Power BI dashboard shows **Rajasthan** as the highest state in the **Total Pollutant** visualization because that chart uses the **sum of pollutant_max values** instead of average pollutant values. This demonstrates how different calculation methods can produce different rankings while still representing the same dataset accurately.

The weather analysis further indicates that **temperature, humidity, rainfall, wind speed, visibility, and UV index** have only **weak relationships** with pollutant levels in this dataset. Since the project uses observations from **a single date (31-05-2026)**, weather conditions alone cannot explain the variation in pollution levels. Other factors such as traffic density, industrial emissions, construction activities, and local geographical conditions are likely to have a much greater influence. Overall, these findings provide a clear picture of pollution distribution across India and form the basis for the recommendations and conclusions presented later in the report.

Rank	City	Average Pollutant
1	Angul	84.14
2	Jaisalmer	75.00
3	Faridabad	66.57
4	Bikaner	64.57
5	Bhiwani	62.43
6	Mandideep	61.57
7	Gummidipoondi	60.40
8	Khora	58.71
9	Bathinda	57.80
10	Sonipat	56.86

21. Recommendations for AQI Improvement

Based on the findings from the AQI analysis, several practical recommendations can be made to improve air quality and reduce pollution levels. These recommendations focus on controlling the major pollutants identified in the project, improving environmental management, and protecting public health. If implemented effectively, these measures can help create cleaner air and a healthier environment for people living in highly polluted areas.

Key Recommendations

- **Reduce PM10 and PM2.5 emissions** by controlling major pollution sources such as road dust, construction activities, industrial smoke, and vehicle exhaust. Regular road cleaning, covering construction materials, installing dust-control systems, and enforcing industrial emission standards can significantly reduce particulate matter pollution.
- **Give priority to high-pollution cities** such as **Angul, Jaisalmer, Faridabad, Bikaner, and Bhiwani**. These cities recorded some of the highest pollution levels in the analysis and should receive more frequent monitoring, stricter pollution-control measures, and targeted environmental improvement programs.
- **Increase green cover** by planting more trees and developing green belts near roads, industrial zones, residential areas, and public spaces. Trees help absorb certain pollutants, reduce dust, improve air quality, and lower urban temperatures.
- **Promote clean and sustainable transportation** by encouraging the use of electric vehicles, public transportation, cycling, and walking. Improving public transport systems and expanding charging infrastructure can help reduce emissions from private vehicles.
- **Strengthen air quality monitoring systems** by increasing the number of monitoring stations, especially in highly polluted regions. Providing **real-time AQI updates and public alerts** can help citizens take necessary precautions during high-pollution days.
- **Improve industrial pollution control** by ensuring industries follow environmental regulations, install modern pollution-control equipment, and regularly monitor emissions. Strict compliance with emission standards can significantly reduce industrial air pollution.
- **Raise public awareness** through educational campaigns that encourage people to avoid open waste burning, reduce the use of firecrackers, minimize unnecessary vehicle idling, and adopt environmentally friendly practices in their daily lives.
- **Support government policies and environmental programs** that focus on reducing emissions, encouraging renewable energy, improving waste management, and promoting sustainable urban development. Long-term policy implementation is essential for maintaining better air quality.
- **Conduct regular air quality studies** using updated datasets collected over longer periods. Continuous monitoring and periodic analysis will help identify pollution trends, evaluate the effectiveness of pollution-control measures, and support better environmental planning.

Expected Benefits

Implementing these recommendations can provide several long-term benefits:

- Improved air quality and healthier living conditions.
- Reduced levels of harmful pollutants such as **PM10** and **PM2.5**.
- Lower risk of respiratory and cardiovascular diseases among the population.
- Better environmental planning and resource allocation by government agencies.
- Increased public awareness and participation in pollution-control efforts.
- More reliable environmental monitoring and faster response during high-pollution events.
- Support for sustainable urban development and long-term environmental protection.

22. Limitations and Assumptions

Every data analytics project has certain limitations and assumptions that should be clearly explained. Mentioning these limitations makes the project more transparent, realistic, and professional. It also helps readers understand the scope of the analysis and prevents incorrect interpretation of the results. While this project provides valuable insights into air quality and weather conditions, the findings should be understood within the boundaries of the available data.

The most important limitation of this project is that the analysis is based primarily on data collected for **31-05-2026**. Since the datasets represent observations from a single day, the project performs a **cross-sectional analysis**, comparing pollution levels across different cities and states on that specific date. Therefore, the results should **not** be interpreted as long-term pollution trends or future predictions.

The weather analysis also shows only **weak correlations** with pollutant levels. These statistical relationships indicate association rather than direct cause-and-effect. Air pollution is influenced by many factors working together, including traffic, industrial activities, construction work, fuel usage, geographical conditions, and weather patterns. Therefore, weather variables alone cannot fully explain changes in pollution levels.

The Power BI dashboard uses different measures such as **pollutant_avg**, **pollutant_min**, and **pollutant_max** for different visualizations. Because each measure represents a different aspect of pollution, some charts may produce different rankings for cities or states. These differences are expected and have been clearly explained throughout the report.

The number of records available for each city depends on the **coverage of monitoring stations**. Cities with more monitoring stations may have more observations than cities with fewer stations. As a result, comparisons between locations should be interpreted carefully, considering differences in data availability.

In addition, the project uses only the AQI, Weather, and Location datasets. It does **not** include other important factors such as **population density, traffic volume, industrial activity, land use, seasonal variations, festivals, or agricultural burning**, all of which can significantly influence air pollution levels. Including these additional datasets in future studies would provide a more comprehensive understanding of air quality.

Project Limitations

- The dataset is **cross-sectional**, comparing multiple cities and states on **31-05-2026** rather than analyzing pollution over several months or years.
- The project is suitable for **location-based comparison** but not for **long-term trend analysis or forecasting**.
- Weather variables show **weak statistical relationships** with pollutant levels and should not be interpreted as direct causes of pollution.
- Different dashboard visuals use different measures (**pollutant_avg**, **pollutant_min**, and **pollutant_max**), which can lead to different rankings while still representing the data correctly.
- Data availability depends on the number of **air quality monitoring stations**, so some cities may have more observations than others.
- The project does not include external factors such as **population, traffic, industrial emissions, seasonal effects, or geographical conditions**, which could improve the interpretation of pollution patterns.
- The findings are based only on the available datasets and should be considered within the scope of the collected data.

Future Scope

To make the project more comprehensive in the future, additional improvements can be made, such as:

- Collecting **daily, monthly, or yearly AQI data** for long-term trend analysis.
- Including **population, traffic, industrial, and land-use data** to better understand the causes of pollution.
- Adding **health-related datasets** to study the impact of air pollution on public health.
- Using **machine learning models** to predict future AQI levels.
- Integrating **real-time AQI APIs** to build live dashboards for continuous environmental monitoring.

23. Future Scope

The Air Quality Index (AQI) Analysis project provides valuable insights into pollution levels across different cities and states. However, there are several opportunities to further improve the project and make it more useful for environmental monitoring and decision-making. By expanding the dataset, incorporating additional data sources, and using advanced analytics techniques, the project can evolve from a descriptive analysis into a predictive and real-time monitoring system.

Currently, the project analyses data collected for **31-05-2026**, which makes it suitable for comparing pollution levels on a single day. In the future, collecting data over longer periods will allow users to study seasonal variations, long-term pollution trends, and changes in air quality over time. This would provide a deeper understanding of how pollution levels change throughout the year. In addition to AQI and weather data, future versions of the project can include external factors such as **population density, traffic volume, industrial activity, construction work, land use, festival periods, crop burning, and geographical conditions**. These additional datasets would help explain the causes of pollution more accurately and support better environmental planning.

The project can also be enhanced by applying **Machine Learning** techniques to build forecasting models. These models can predict future AQI values and pollutant concentrations, allowing government agencies and environmental organizations to take preventive measures before pollution levels become hazardous.

Power BI dashboards can also be made more interactive by including **geographical map visualizations**, allowing users to view pollution levels across cities and monitoring stations directly on a map. This would make it easier to identify pollution hotspots and regional pollution patterns.

Another useful improvement would be to automate the complete data pipeline. Instead of manually importing new CSV files, the dashboard could automatically refresh whenever new AQI and weather data becomes available. This would make the system more suitable for continuous environmental monitoring.

Finally, an **AQI alert system** could be developed to notify authorities or users whenever pollution levels exceed predefined safety limits. Such a feature would support faster decision-making and help protect public health during high-pollution events.

Future Enhancements

- **Collect daily, monthly, and yearly AQI data** to study seasonal variations and long-term pollution trends.
- **Include additional datasets** such as population density, traffic flow, industrial zones, construction activities, crop burning, festivals, and land-use information for more comprehensive analysis.
- **Develop Machine Learning models** to predict future AQI values and pollutant levels, enabling early warning and better planning.
- **Create interactive map visualizations** in Power BI to display pollution levels by city, district, and monitoring station.
- **Automate data collection and dashboard refresh**, allowing the system to update automatically whenever new AQI and weather datasets are available.
- **Build an air quality alert system** that sends notifications when pollutant levels exceed safe thresholds, helping authorities and citizens take timely action.
- **Integrate real-time AQI APIs** so the dashboard displays live pollution data instead of relying only on historical datasets.
- **Add health impact analysis** by combining hospital records, respiratory disease statistics, and AQI data to study the relationship between pollution and public health.
- **Develop a web or mobile application** where users can check live AQI, weather conditions, pollution forecasts, and health recommendations for their location.
- **Use Artificial Intelligence (AI) and Generative AI** to automatically generate pollution reports, summarize trends, and provide recommendations for government agencies and environmental organizations.

24. Conclusion

The **Air Quality Index (AQI) Analysis Project** successfully demonstrates a complete **end-to-end data analytics workflow**, covering every stage of the data analysis process. The project begins with collecting and preparing three structured datasets, followed by data cleaning in Excel, database management in PostgreSQL, exploratory data analysis using Python, interactive dashboard development in Power BI, and finally presenting the results through PowerPoint and this comprehensive PDF report. This workflow shows how multiple analytical tools can be integrated to transform raw environmental data into meaningful insights that support better decision-making.

The analysis identified several important pollution patterns across different cities, states, and pollutant types. Among all the pollutants, **PM10** recorded the highest average pollutant value, followed by **PM2.5**, making particulate matter the most significant contributor to poor air quality in this dataset. Since these pollutants can have serious effects on human health, they should remain the primary focus of future pollution-control initiatives.

The city-level analysis highlighted **Angul, Jaisalmer, Faridabad, Bikaner, and Bhiwani** as the locations with the highest average pollution levels. These cities may require increased environmental monitoring, stricter pollution-control measures, and targeted government action to improve air quality. At the state level, **Haryana** recorded the highest **average pollutant value**, while **Rajasthan** ranked highest in the Power BI dashboard when pollution was measured using the **sum of pollutant_max values**. These results demonstrate how different performance measures provide different but equally valuable perspectives on the same dataset.

The weather analysis added useful environmental context by comparing pollutant levels with variables such as **temperature, humidity, rainfall, wind speed, visibility, and UV index**. However, the correlation analysis showed that these relationships are generally **weak**, indicating that weather conditions alone do not explain changes in pollution levels. Air pollution is influenced by a combination of factors, including **vehicle emissions, industrial activities, construction dust, population density, geographical conditions, and weather patterns**, all of which contribute together to the overall air quality.

Since the project is based on observations collected for **31-05-2026**, it provides a **cross-sectional comparison** of pollution levels across different locations rather than a long-term trend analysis. Nevertheless, the project successfully demonstrates how data analytics techniques can be used to identify pollution hotspots, compare environmental conditions, and generate actionable insights from real-world datasets.

Overall, the project illustrates the practical application of **Excel, PostgreSQL, Python, and Power BI** in solving an environmental problem through data analytics. It not only strengthens technical skills in data cleaning, database management, SQL querying, exploratory data analysis, visualization, and dashboard development but also highlights how data-driven insights can support **environmental planning, public health awareness, and informed decision-making**. The project provides a strong foundation for future enhancements such as real-time monitoring, predictive analytics, and AI-based pollution forecasting, making it a valuable example of an end-to-end data analytics solution.

25. References

The report is based on the files supplied inside the AQI Analysis Project ZIP package and the reference format of the Mental Health Survey Analysis project report.

KEY POINTS

- **AQI DATASET.csv** : main pollutant dataset used for EDA and dashboard visuals.
- **LOCATION DATASET.csv** : city and state master dataset.
- **WEATHER DATASET.csv** : weather dataset used for weather impact analysis.
- **AQI ANALYSIS EDA.ipynb** : Python notebook with pandas, numpy, matplotlib, and seaborn analysis.
- **AQI DATA QUERY.sql, LOCATION DATA QUERY.sql, WEATHER DATA QUERY.sql** : PostgreSQL table and query scripts.
- **AQI ANALYSIS DASHBOARD.pbix** : Power BI dashboard file.
- **Dashboard 1.png and Dashboard 2.png** : exported Power BI dashboard screenshots.
- **AQI ANALYSIS PPT.pptx** : project presentation and report summary.

26. Appendix: Final Project Checklist

This appendix summarizes the deliverables and confirms that the report covers the complete project path from datasets to Python EDA, Power BI dashboard, and PPT report.

KEY POINTS

- Three core datasets explained: Location, AQI, and Weather.
- PostgreSQL tables and SQL analysis documented.
- Python EDA workflow and all major chart insights included.
- Power BI dashboard pages and visual interpretation included.
- PPT report role and final recommendations included.
- Final report generated as a 50-page downloadable PDF.

Dataset	Columns
Location	Location_ID, state, city
AQI	AQI_ID, Location_ID, country, date, state, city, station, latitude, longitude, pollutant id, pollutant min, pollutant max, pollutant avg
Weather	Weather_ID, Location_ID, date, city, state, latitude, longitude, temperature max in C, temperature min in C, temperature avg in C, humidity Pct, rainfall in mm, wind speed in km/h, wind direction, pressure in h/Pa, visibility in km, cloud cover Pct, uv index